

UNIVERSITY PHYSICS I - CLASSICAL MECHANICS



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Preface

Students: if this is too long, at the very least read the last four paragraphs.

Thank you! For many years Eric Mazur's Principles and Practice of Physics was the required textbook for University Physics I at the University of Arkansas. In writing this open-source replacement I have tried to preserve some of its best features, while at the same time condensing much of the presentation, and reworking several sections that did not quite fit the needs of our curriculum: primarily, the chapters on Thermodynamics, Waves, and Work. I have also skipped entirely the chapter on the "Principle of Relativity," and instead distributed its contents among other chapters: in particular, the Galilean reference frame transformations are now introduced at the very beginning of the book, as are the law of inertia and the concept of inertial reference frames.

Over the past few decades, there has been a trend to increase the size of introductory physics textbooks, by including more and more visual aids (pictures, diagrams, boxes...), as well as lengthier and more detailed explanations, perhaps in an attempt to reach as many students as possible, and maybe even to take the place of the instructor altogether. It seems to me that the result is rather the opposite: a massive (and expensive) tome that no student could reasonably be expected to read all the way through, at a time when "TL;DR" has become a popular acronym, and visual learning aids (videotaped lectures, demonstrations, and computer simulations) are freely available everywhere.

Our approach at the University of Arkansas, developed as a result of the work of Physics Education Research experts John and Gay Stewart, is based instead on two essential facts. First, that different students learn differently: some will learn best from a textbook, others will learn best from a lecture, and most will only really learn from a hands-on approach, by working out the answers to questions themselves. Second, just about everybody will benefit from repeated presentations of the material to be learned, in different environments and even from slightly different points of view.

In keeping with this, we start by requiring the students to read the textbook material before coming to lecture, and also take an online "reading quiz" where they can check their understanding of what they have read. Then, in the lecture, they will have an opportunity to see the material presented again, as a sort of executive summary delivered by, typically, a different instructor, who will also be able to answer any questions they might have about the book's presentation. Additionally, the instructor will directly test the students' understanding by means of conceptual questions asked of the whole class, which are to be answered using clickers. This prompts the students to think harder about the material, and encourages them to discuss it on the spot with their classmates.

Immediately after each lecture, the students will have a lab activity where they will be able to verify experimentally the concepts and principles to which they have just been introduced. Finally, every week they will have an "open response" homework assignment where they get to apply the principles, mathematically, to concrete problems. For both the labs and the homework, additional assistance is provided by a group of dedicated teaching assistants, who are often able to explain the material to the students in a way that better relates to their own experience.

In all this, the textbook is expected to play an important role, but certainly not to be the students' only (nor even, necessarily, the primary) source of understanding or knowledge. Its job is to start the learning process, and to stand by to provide a reference (among possibly several others) afterwards. To fulfill this role, perhaps the most essential requirement is that it should be readable, and hence concise enough for every reading assignment to be of manageable size. This (as well as a sensible organization, clear explanations, and a minimal assortment of worked-out exercises and end-of chapter problems) is what I have primarily tried to provide here. A student who wants more information than provided in this textbook, or alternative explanations, or more worked-out examples, can certainly get these from many other sources: first, of course, the instructor and the teaching assistants, whose essential role as learning facilitators has to be recognized from the start. Then, there is a variety of alternative textbooks available: the best choice, probably, would still be Mazur's Principles and Practice of Physics, since it uses the same terminology and notation, introduces the material in almost the same sequence, and has tons of worked-out examples and self-quiz conceptual questions. That book is available on reserve in the Physics library, where it can be consulted by anybody. If a student feels the need for an alternative textbook that they can actually take home, one option is, of course, to actually buy Mazur's (which is what everybody had to do before); another option is to explore other open-source textbooks available online, which have been around longer than this one and benefit from more worked-out exercises and a more conventional presentation. One such book is [University Physics I](#).

Finally, there are also a large number of other online resources, although I would advise the students to approach them with caution, since not all of them may be totally rigorous, and some may end up being more confusing than helpful. Some of my

students have found the Khan Academy lectures and/or the “Flipping Physics” lectures helpful; I personally would recommend the lectures of Walter Lewin at MIT, if only for the wide array of cool demonstrations you can see there.

One last word, for the students who may have read this far, concerning the use of equations and “proofs” in this book. It is essential to the nature of physics to be able to cast its results in mathematical terms, and to use math to explain and predict new results; hence, equations and mathematical derivations are integral parts of any physics textbook. I have, however, tried to keep the math as simple as possible throughout, and I would not want a lengthy mathematical derivation to get in the way of your reading assignment. If you are reading the text and come upon several lines of math, skim them at first to see if they make sense, but if you get stuck do not spend too much time on them: move on to the bottom line, and keep reading from there. You can always ask your instructor or TA for help with the math later.

I would, however, encourage you to return, eventually, to any bit of math that you found challenging the first time around. Do try to go through all the algebra yourself! I have occasionally skipped intermediate steps, just to keep the math from overwhelming the text: but these are typically straightforward manipulations (multiplying or dividing both sides by something, moving something from one side of the equation to the other, multiplying out a parenthesis or, conversely, pulling out a common factor or denominator...). If you actually work out, on your own, all the missing steps, you will find it’s a great way to improve your algebra skills. This is something that will make it much easier for you to deal with the homework and the exams later on this semester—and for the rest of your career as well.

CHAPTER OVERVIEW

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1.1: Introduction

Classical mechanics is the branch of physics that deals with the study of the motion of anything (roughly speaking) larger than an atom or a molecule. That is a lot of territory, and the methods and concepts of classical mechanics are at the foundation of any branch of science or engineering that is concerned with the motion of anything from a star to an amoeba—fluids, rocks, animals, planets, and any and all kinds of machines. Moreover, even though the accurate description of processes at the atomic level requires the (formally very different) methods of quantum mechanics, at least three of the basic concepts of classical mechanics that we are going to study this semester, namely, momentum, energy, and angular momentum, carry over into quantum mechanics as well, with the last two playing, in fact, an essential role.

Particles in Classical Mechanics

In the study of motion, the most basic starting point is the concept of the *position* of an object. Clearly, if we want to describe accurately the position of a macroscopic object such as a car, we may need a lot of information, including the precise shape of the car, whether it is turned this way or that way, and so on; however, if all we want to know is how far the car is from Fort Smith or Fayetteville, we do not need any of that: we can just treat the car as a dot, or mathematical point, on the map—which is the way your GPS screen will show it, anyway. When we do this, we say that we are describing the car (or whatever the macroscopic object may be) as a **particle**.

In classical mechanics, an “ideal” particle is an object with no appreciable size—a mathematical point. In one dimension (that is to say, along a straight line), its position can be specified just by giving a single number, the distance from some reference point, as we shall see in a moment (in three dimensions, of course, three numbers are required). In terms of energy (which is perhaps the most important concept in all of physics, and which we will introduce properly in due course), an ideal particle has only one kind of energy, what we will later call *translational kinetic energy*; it cannot have, for instance, rotational kinetic energy (since it has “no shape” for practical purposes), or any form of internal energy (elastic, thermal, etc.), since we assume it is too small to have any internal structure in the first place.

The reason this is a useful concept is not just that we can often treat extended objects as particles in an approximate way (like the car in the example above), but also, and most importantly, that if we want to be more precise in our calculations, *we can always treat an extended object (mathematically) as a collection of “particles.”* The physical properties of the object, such as its energy, momentum, rotational inertia, and so forth, can then be obtained by adding up the corresponding quantities for all the particles making up the object. Not only that, but the interactions between two extended objects can also be calculated by adding up the interactions between all the particles making up the two objects. This is how, once we know the form of the gravitational force between two particles (which is fairly simple, as we will see in [Chapter 10](#)), we can use that to calculate the force of gravity between a planet and its satellites, which can be fairly complicated in detail, depending, for instance, on the relative orientation of the planet and the satellite.

The mathematical tool we use to calculate these “sums” is *calculus*—specifically, integration—and you will see many examples of this... in your calculus courses. Calculus I is only a corequisite for this course, so we will not make a lot of use of it here, and in any case you would need multidimensional integrals, which are an even more advanced subject, to do these kinds of calculations. But it may be good for you to keep these ideas on the back of your mind. Calculus was, in fact, invented by Sir Isaac Newton precisely for this purpose, and the developments of physics and mathematics have been closely linked together ever since.

Anyway, back to particles, the plan for this semester is as follows: we will start our description of motion by treating every object (even fairly large ones, such as cars) as a “particle,” because we will only be concerned at first with its translational motion and the corresponding energy. Then we will progressively make things more complex: by considering systems of two or more particles, we will start to deal with the *internal energy* of a system. Then we will move to the study of *rigid bodies*, which are another important idealization: extended objects whose parts all move together as the object undergoes a translation or a rotation. This will allow us to introduce the concept of rotational kinetic energy. Eventually we will consider *wave motion*, where different parts of an extended object (or “medium”) move relative to each other. So, you see, there is a logical progression here, with most parts of the course building on top of the previous ones, and energy as one of the main connecting themes.

Aside- The Atomic Perspective

As an aside, it should perhaps be mentioned that the building up of classical mechanics around this concept of ideal particles had nothing to do, initially, with any belief in “atoms,” or an atomic theory of matter. Indeed, for most 18th and 19th century physicists,

matter was supposed to be a continuous medium, and its (mental) division into particles was just a mathematical convenience.

The atomic hypothesis became increasingly more plausible as the 19th century wore on, and by the 1920's, when quantum mechanics came along, physicists had to face a surprising development: matter, it turned out, was indeed made up of “elementary particles,” but these particles could not, in fact, be themselves described by the laws of classical mechanics. One could not, for instance, attribute to them simultaneously well-defined positions and velocities. Yet, in spite of this, most of the conclusions of classical mechanics remain valid for macroscopic objects, because, most of the time, it is OK to (formally) “break up” extended objects into chunks that are small enough to be treated as particles, but large enough that one does not need quantum mechanics to describe their behavior.

Quantum properties were first found to manifest themselves at the macroscopic level when dealing with thermal energy, because at one point it really became necessary to figure out where and how the energy was stored at the truly microscopic (atomic) level. Thus, after centuries of successes, classical mechanics met its first failure with the so-called *problem of the specific heats*, and a completely new physical theory—quantum mechanics—had to be developed in order to deal with the newly-discovered atomic world. But all this, as they say, is another story, and for our very brief dealings with thermal physics—the last chapter in this book—we will just take specific heats as given, that is to say, something you measure (or look up in a table), rather than something you try to calculate from theory.

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1.2: Position, Displacement, Velocity

Kinematics is the part of mechanics that deals with the mathematical description of motion, leaving aside the question of what causes an object to move in a certain way. Kinematics, therefore, does not include such things as forces or energy, which fall instead under the heading of dynamics. It may be said, then, that kinematics by itself is not true physics, but only applied mathematics; yet it is still an essential part of classical mechanics, and its most natural starting point. This chapter (and parts of the next one) will introduce the basic concepts and methods of kinematics in one dimension.

Position

As stated in the previous section, we are initially interested only in describing the motion of a “particle,” which can be thought of as a mathematical point in space. (Later on we will see that, even for an extended object or system, it is often useful to consider the motion of a specific point that we call the system’s center of mass.) A point in three dimensions can be located by giving three numbers, known as its *Cartesian coordinates* (or, more simply, its *coordinates*). In two dimensions, this works as shown in Figure 1.2.1 below. As you can see, the coordinates of a point just tell us how to find it by first moving a certain distance x , from a previously-agreed origin, along a horizontal (or x) axis, and then a certain distance y along a vertical (or y) axis. (Or, of course, you could equally well first move vertically and then horizontally.)

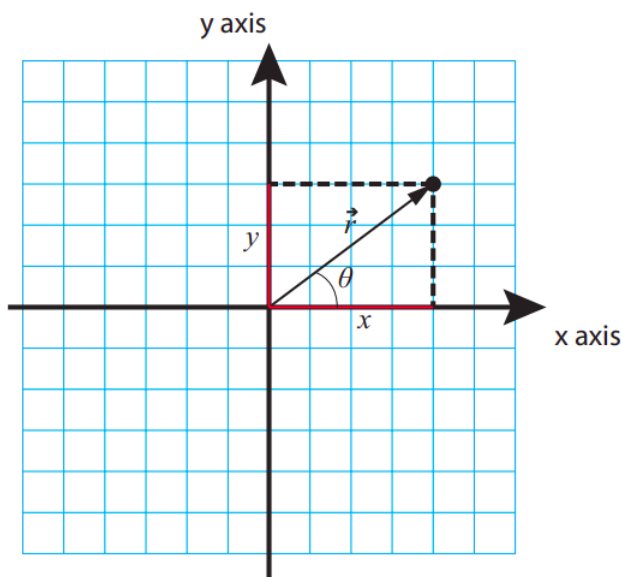


Figure 1.2.1: The position vector, $\vec{r}$, of a point, and its x and y components (the point’s coordinates).

The quantities x and y are taken to be positive or negative depending on what side of the origin the point is on. Typically, we will always start by choosing a *positive direction* for each axis, as the direction along which the algebraic value of the corresponding coordinate increases. This is often chosen to be to the right for the horizontal axis, and upwards for the vertical axis, but there is nothing that says we cannot choose a different convention if it turns out to be more convenient. In Figure 1.2.1, the arrows on the axes denote the positive direction for each. Going by the grid, the coordinates of the point shown are $x = 4$ units, $y = 3$ units.

In two or three dimensions (and even, in a sense, in one dimension), the coordinates of a point can be interpreted as the *components of a vector* that we call the point’s **position vector**, and denote by $\vec{r}$ (sometimes boldface letters are used for vectors, instead of an arrow on top; in that case, the position vector would be denoted by $\mathbf{r}$). A **vector** is a mathematical object, with specific geometric and algebraic properties, that physicists use to represent a quantity that has both a magnitude and a direction. The *magnitude* of the position vector in Figure 1.2.1 is just the length of the arrow, which is to say, 5 length units (by the Pythagorean theorem, the length of $\vec{r}$, which we will often write using absolute value bars as $|\vec{r}|$, is equal to $\sqrt{x^2 + y^2}$); this is just the straight-line distance of the point to the origin. The *direction* of $\vec{r}$, on the other hand, can be specified in a number of ways; a common convention is to give the value of the angle that it makes with the positive x axis, which I have denoted in the figure as θ (in this case, you can verify that $\theta = \tan^{-1}(y/x) = 36.9^\circ$). In three dimensions, two angles would be needed to completely specify the direction of $\vec{r}$.

As you can see, giving the magnitude and direction of $\vec{r}$ is a way to locate the point that is completely equivalent to giving its coordinates x and y . By the same token, the coordinates x and y are a way to specify the vector $\vec{r}$ that is completely equivalent to giving its magnitude and direction. As I stated above, we call x and y the components (or sometimes, to be more specific, the Cartesian components) of the vector $\vec{r}$. In a sense all the vectors that will be introduced later on this semester will derive their geometric and algebraic properties from the position vector $\vec{r}$, so once you know how to deal with one vector, you can deal with them all. The geometric properties (by which I mean, how to relate a vector's components to its magnitude and direction) I have just covered, and will come back to later on in this chapter, and again in [Chapter 8](#); the algebraic properties (how to add vectors and multiply them by ordinary numbers, which are called *scalars* in this context) I will introduce along the way.

For the first few chapters this semester, we are going to be primarily concerned with motion in one dimension (that is to say, along a straight line, backwards or forwards), in which case all we need to locate a point is one number, its x (or y , or z) coordinate; we do not then need to worry particularly about vector algebra. Alternatively, we can simply say that a vector in one dimension is essentially the same as its only component, which is just a positive or negative number (the magnitude of the number being the magnitude of the vector, and its sign indicating its direction), and has the algebraic properties that follow naturally from that.

The description of the motion that we are aiming for is to find a *function of time*, which we denote by $x(t)$, that gives us the point's position (that is to say, the value of x) for any value of the time parameter, t . (See Equation (1.2.10), below, for an example.) Remember that x stands for a number that can be positive or negative (depending on the side of the origin the point is on), and has dimensions of length, so when giving a numerical value for it you must always include the appropriate units (meters, centimeters, miles...). Similarly, t stands for the time elapsed since some more or less arbitrary “origin of time,” or time zero. Normally t should always be positive, but in special cases it may make sense to consider negative times (think of how we count years: “AD” would correspond to “positive” and “BC” would correspond to negative—the difference being that there is actually no year zero!). Anyway, t also is a number with dimensions, and must be reported with its appropriate units: seconds, minutes, hours, etc.

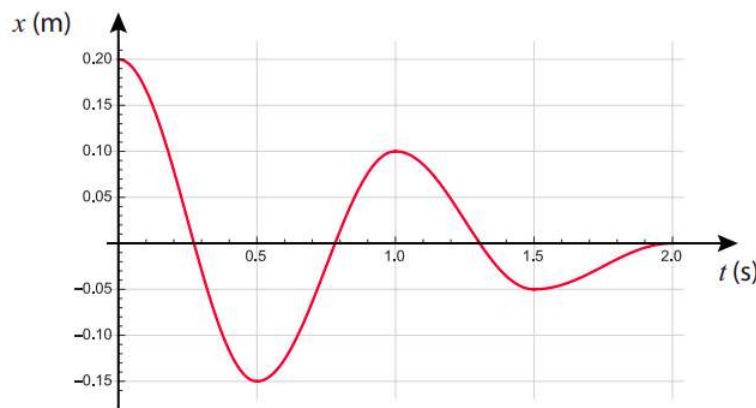


Figure 1.2.2: A possible position vs. time graph for an object moving in one dimension

We will be often interested in plotting the position of an object as a function of time—that is to say, the graph of the function $x(t)$. This may, in principle, have any shape, as you can see in Figure 1.2.2 above. In the lab, you will have a chance to use a position sensor that will automatically generate graphs like that for you on the computer, for any moving object that you aim the position sensor at. It is, therefore, important that you learn how to “read” such graphs. For example, Figure 1.2.2 shows an object that starts, at the time $t = 0$, a distance 0.2 m away and to the right of the origin (so $x(0) = 0.2$ m), then moves in the negative direction to $x = -0.15$ m, which it reaches at $t = 0.5$ s; then turns back and moves in the opposite direction until it reaches the point $x = 0.1$ m, turns again, and so on. Physically, this could be tracking the damped oscillations of a system such as an object attached to a spring and sliding over a surface that exerts a friction force on it (see Example 11.5.1).

Displacement

In one dimension, the **displacement** of an object over a given time interval is a quantity that we denote as Δx , and equals the difference between the object's initial and final positions (in one dimension, we will often call the “position coordinate” simply the “position,” for short):

$$\Delta x = x_f - x_i \quad (1.2.1)$$

Here the subscript i denotes the object's position at the beginning of the time interval considered, and the subscript f its position at the end of the interval. The symbol Δ will consistently be used throughout this book to denote a *change* in the quantity following the symbol, meaning the difference between its initial value and its final value. The time interval itself will be written as Δt and can be expressed as

$$\Delta t = t_f - t_i \quad (1.2.2)$$

where again t_i and t_f are the initial and final values of the time parameter (imagine, for instance, that you are reading time in seconds on a digital clock, and you are interested in the change in the object's position between second 130 and second 132: then $t_i = 130$ s, $t_f = 132$ s, and $\Delta t = 2$ s).

You can practice reading off displacements from Figure 1.2.2. The displacement between $t_i = 0.5$ s and $t_f = 1$ s, for instance, is 0.25 m ($x_i = -0.15$ m, $x_f = 0.1$ m). On the other hand, between $t_i = 1$ s and $t_f = 1.3$ s, the displacement is $\Delta x = 0 - 0.1 = -0.1$ m

Notice two important things about the displacement. First, it can be positive or negative. Positive means the object moved, overall, in the positive direction; negative means it moved, overall, in the negative direction. Second, even when it is positive, the displacement does not always equal the distance traveled by the object (distance, of course, is always defined as a positive quantity), because if the object “doubles back” on its tracks for some distance, that distance does not count towards the overall displacement. For instance, looking again at Figure 1.2.2, in between the times $t_i = 0.5$ s and $t_f = 1.5$ s the object moved first 0.25 m in the positive direction, and then 0.15 m in the negative direction, for a total distance traveled of 0.4 m; however, the total displacement was just 0.1 m.

In spite of these quirks, the total displacement is, mathematically, a useful quantity, because often we will have a way (that is to say, an equation) to calculate Δx for a given interval, and then we can rewrite Equation (1.2.1) so that it reads

$$x_f = x_i + \Delta x. \quad (1.2.3)$$

That is to say, if we know where the object started, and we have a way to calculate Δx , we can easily figure out where it ended up. You will see examples of this sort of calculation in the homework later on.

Extension to Two Dimensions

In two dimensions, we write the displacement as the vector

$$\Delta \vec{r} = \vec{r}_f - \vec{r}_i. \quad (1.2.4)$$

The components of this vector are just the differences in the position coordinates of the two points involved; that is, $(\Delta \vec{r})_x$ (a subscript x , y , etc., is a standard way to represent the x , y . . . component of a vector) is equal to $x_f - x_i$, and similarly $(\Delta \vec{r})_y = y_f - y_i$.

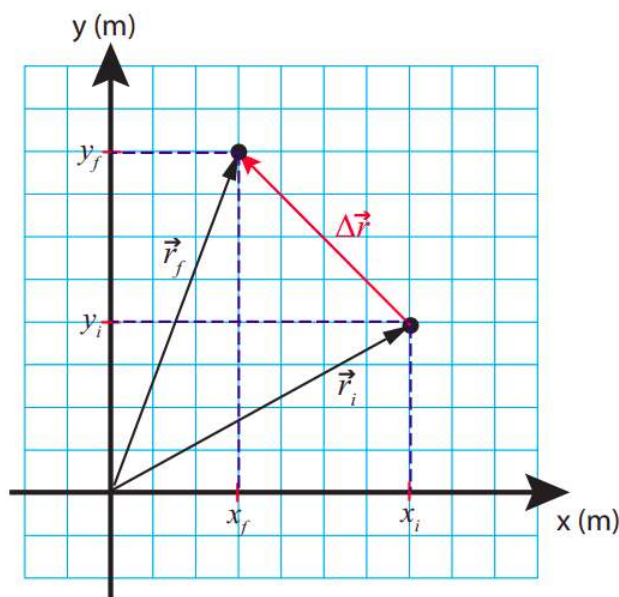


Figure 1.2.3: The displacement vector for a particle that was initially at a point with position vector $\vec{r}_i$ and ended up at a point with position vector $\vec{r}_f$ is the *difference* of the position vectors.

Figure 1.2.3 shows how this makes sense. The x component of $\Delta\vec{r}$ in the figure is $\Delta x = 3 - 7 = -4$ m; the y component is $\Delta y = 8 - 4 = 4$ m. This basically shows you how to subtract (and, by extension, add, since $\vec{r}_f = \vec{r}_i + \Delta\vec{r}$) vectors: you just subtract (or add) the corresponding components. Note how, by the Pythagorean theorem, the length (or magnitude) of the displacement vector, $|\Delta\vec{r}| = \sqrt{(x_f - x_i)^2 + (y_f - y_i)^2}$, equals the straight-line distance between the initial point and the final point, just as in one dimension; of course, the particle could have actually followed a very different path from the initial to the final point, and therefore traveled a different distance.

Velocity

Average Velocity

If you drive from Fayetteville to Fort Smith in 50 minutes, your average speed for the trip is calculated by dividing the distance of 59.2 mi by the time interval:

$$\text{average speed} = \frac{\text{distance}}{\Delta t} = \frac{59.2 \text{ mi}}{50 \text{ min}} = \frac{59.2 \text{ mi}}{50 \text{ min}} \times \frac{60 \text{ min}}{1 \text{ hr}} = 71.0 \text{ mph} \quad (1.2.5)$$

(this equation, incidentally, also shows you how to convert units, and how you will be expected to work with significant figures this semester: the rule of thumb is, keep four significant figures in all intermediate calculations, and report three in the final result).

The way we define *average velocity* is similar to average speed, but with one important difference: we use the *displacement*, instead of the distance. So, the average velocity v_{av} of an object, moving along a straight line, over a time interval Δt is

$$v_{av} = \frac{\Delta x}{\Delta t}. \quad (1.2.6)$$

This definition has all the advantages and the quirks of the displacement itself. On the one hand, it automatically comes with a sign (the same sign as the displacement, since Δt will always be positive), which tells us in what direction we have been traveling. On the other hand, it may not be an accurate estimate of our average *speed*, if we doubled back at all. In the most extreme case, for a roundtrip (leave Fayetteville and return to Fayetteville), the average velocity would be zero, since $x_f = x_i$ and therefore $\Delta x = 0$.

It is clear that this concept is not going to be very useful in general, if the object we are tracking has a chance to double back in the time interval Δt . A way to prevent this from happening, and also getting a more meaningful estimate of the object's speed at any instant, is to make the time interval very small. This leads to a new concept, that of *instantaneous velocity*.

Instantaneous Velocity

We define the instantaneous velocity of an object (a “particle”), at the time $t = t_i$, as the mathematical limit

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}. \quad (1.2.7)$$

The meaning of this is the following. Suppose we compute the ratio $\Delta x/\Delta t$ over successively smaller time intervals Δt (all of them starting at the same time t_i). For instance, we can start by making $t_f = t_i + 1$ s, then try $t_f = t_i + 0.5$ s, then $t_f = t_i + 0.1$ s, and so on. Naturally, as the time interval becomes smaller, the corresponding displacement will also become smaller—the particle has less and less time to move away from its initial position, x_i . The hope is that the successive ratios $\Delta x/\Delta t$ will *converge* to a definite value: that is to say, that at some point we will start getting very similar values, and that beyond a certain point making Δt any smaller will not change any of the significant digits of the result that we care about. This limit value is the *instantaneous velocity* of the object at the time t_i .

When you think about it, there is something almost a bit self-contradictory about the concept of instantaneous velocity. You cannot (in practice) determine the velocity of an object if all you are given is a literal instant. You cannot even tell if the object is moving, if all you have is one instant! Motion requires more than one instant, the passage of time. In fact, all the “instantaneous” velocities that we can measure, with any instrument, are always really average velocities, only the average is taken over very short time intervals. Nevertheless, the fact is that for any reasonably well-behaved position function $x(t)$, the limit in Equation (1.2.7) is *mathematically* well-defined, and it equals what we call, in calculus, the *derivative* of the function $x(t)$:

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}. \quad (1.2.8)$$

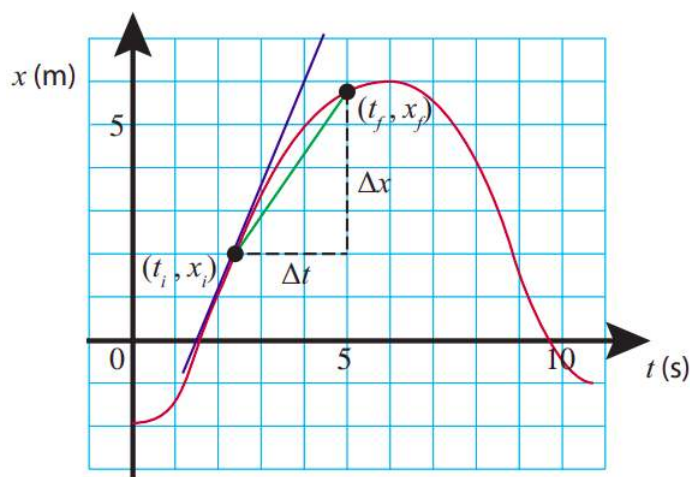


Figure 1.2.4: The slope of the green segment is the average velocity for the time interval Δt shown. As Δt becomes smaller, this approaches the slope of the tangent at the point (t_i, x_i)

In fact, there is a nice geometric interpretation for this quantity: namely, it is the slope of a line tangent to the x -vs- t curve at the point (t_i, x_i) . As Figure 1.2.4 shows, the average velocity $\Delta x/\Delta t$ is the slope (rise over run) of a line segment drawn from the point (t_i, x_i) to the point (t_f, x_f) (the green line in the figure). As we make the time interval smaller, by bringing t_f closer to t_i (and hence, also, x_f closer to x_i), the slope of this segment will approach the slope of the tangent line at (t_i, x_i) (the blue line), and this will be, by the definition (1.2.7), the instantaneous velocity at that point.

This geometric interpretation makes it easy to get a qualitative feeling, from the position-vs-time graph, for when the particle is moving more or less fast. A large slope means a steep rise or fall, and that is when the velocity will be largest—in magnitude. A steep rise means a large positive velocity, whereas a steep drop means a large negative velocity, by which I mean a velocity that is given by a negative number which is large in absolute value. In the future, to simplify sentences like this one, I will just use the word “speed” to refer to the magnitude (that is to say, the absolute value) of the instantaneous velocity. Thus, speed (like distance) is always a positive number, by definition, whereas velocity can be positive or negative; and a steep slope (positive or negative) means the speed is large there.

Conversely, looking at the sample x -vs- t graphs in this chapter, you may notice that there are times when the tangent is horizontal, meaning it has zero slope, and so the instantaneous velocity at those times is zero (for instance, at the time $t = 1.0$ s in Figure 1.2.2). This makes sense when you think of what the particle is actually doing at those special times: it is just changing direction, so its velocity is going, for instance, from positive to negative. The way this happens is, it slows down, down... the velocity gets smaller and smaller, and then, for just an instant (literally, a mathematical point in time), it becomes zero before, the next instant, going negative.

We will be coming back to this “reading of graphs” in the lab and the homework, as well as in the next chapter, when we introduce the concept of acceleration.

Motion With Constant Velocity

If the instantaneous velocity of an object never changes, it means that it is always moving in the same direction with the same speed. In that case, the instantaneous velocity and the average velocity coincide, and that means we can write $v = \Delta x / \Delta t$ (where the size of the interval Δt could now be anything), and rewrite this equation in the form

$$\Delta x = v \Delta t \quad (1.2.9)$$

which is the same as

$$x_f - x_i = v(t_f - t_i)$$

Now suppose we keep t_i constant (that is, we fix the initial instant) but allow the time t_f to change, so we will just write t for an arbitrary value of t_f , and x for the corresponding value of x_f . We end up with the equation

$$x - x_i = v(t - t_i)$$

which we can also write as

$$x(t) = x_i + v(t - t_i) \quad (1.2.10)$$

after some rearranging, and where the notation $x(t)$ has been introduced to emphasize that we want to think of x as a function of t . This is, not surprisingly, the equation of a straight line—a “curve” which is its own tangent and always has the same slope.

(Please make sure that you are not confused by the notation in Equation (1.2.10). The parentheses around the t on the left-hand side mean that we are considering the position x as a function of t . On the other hand, the parentheses around the quantity $t - t_i$ on the right-hand side mean that we are multiplying this quantity by v , which is a constant here. This distinction will be particularly important when we introduce the function $v(t)$ next.)

Either one of equations (1.2.9) or (1.2.11) can be used to solve problems involving motion with constant velocity, and again you will see examples of this in the homework.

Motion With Changing Velocity

If the velocity changes with time, obtaining an expression for the position of the object as a function of time may be a nontrivial task. In the next chapter we will study an important special case, namely, when the velocity changes at a constant rate (constant acceleration).

For the most general case, a graphical method that is sometimes useful is the following. Suppose that we know the function $v(t)$, and we graph it, as in Figure 1.2.5 below. Then the area under the curve in between any two instants, say t_i and t_f , is equal to the total displacement of the object over that time interval.

The idea involved is known in calculus as *integration*, and it goes as follows. Suppose that I break

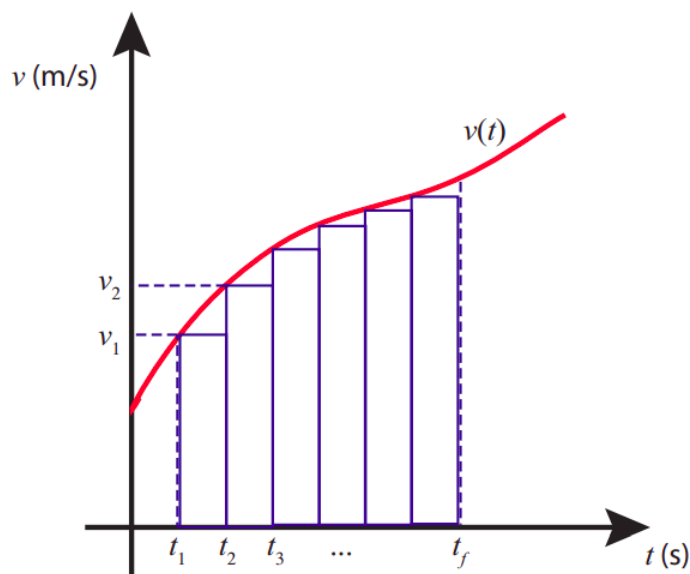


Figure 1.2.5: How to get the displacement from the area under the v -vs- t curve.

down the interval from t_i to t_f into equally spaced subintervals, beginning at the time t_i (which I am, equivalently, going to call t_1 , that is, $t_1 \equiv t_i$, so I have now $t_1, t_2, t_3, \dots, t_f$). Now suppose I treat the object's motion over each subinterval as if it were motion with constant velocity, the velocity being that at the beginning of the subinterval. This, of course, is only an approximation, since the velocity is constantly changing; but, if you look at Figure 1.2.5, you can convince yourself that it will become a better and better approximation as I increase the number of intermediate points and the rectangles shown in the figure become narrower and narrower. In this approximation, the displacement during the first subinterval would be

$$\Delta x_1 = v_1 (t_2 - t_1) \quad (1.2.11)$$

where $v_1 = v(t_1)$; similarly, $\Delta x_2 = v_2(t_3 - t_2)$, and so on.

However, Equation (1.2.11) is just the area of the first rectangle shown under the curve in Figure 1.2.5 (the base of the rectangle has "length" $t_2 - t_1$, and its height is v_1). Similarly for the second rectangle, and so on. So the sum $\Delta x_1 + \Delta x_2 + \dots$ is both an approximation to the area under the v -vs- t curve, and an approximation to the total displacement Δt . As the subdivision becomes finer and finer, and the rectangles narrower and narrower (and more numerous), both approximations become more and more accurate. In the limit of "infinitely many," infinitely narrow rectangles, you get both the total displacement and the area under the curve exactly, and they are both equal to each other. Mathematically, we would write

$$\Delta x = \int_{t_i}^{t_f} v(t) dt \quad (1.2.12)$$

where the stylized "S" (for "sum") on the right-hand side is the symbol of the operation known as *integration* in calculus. This is essentially the inverse of the process known as differentiation, by which we got the velocity function from the position function, back in Equation (1.2.8).

This graphical method to obtain the displacement from the velocity function is sometimes useful, if you can estimate the area under the v -vs- t graph reliably. An important point to keep in mind is that rectangles under the horizontal axis (corresponding to negative velocities) have to be added as having negative area (since the corresponding displacement is negative); see example 1.5.1 at the end of this chapter.

Extension to Two Dimensions

In two (or more) dimensions, you define the average velocity vector as a vector $\vec{v}_{av}$ whose components are $v_{av,x} = \Delta x / \Delta t$, $v_{av,y} = \Delta y / \Delta t$, and so on (where Δx , Δy , ... are the corresponding components of the displacement vector $\Delta \vec{r}$). This can be written equivalently as the single vector equation

$$\vec{v}_{av} = \frac{\Delta \vec{r}}{\Delta t}. \quad (1.2.13)$$

This tells you how to multiply (or divide) a vector by an ordinary number: you just multiply (or divide) each component by that number. Note that, if the number in question is positive, this operation does not change the direction of the vector at all, it just *scales* it up or down (which is why ordinary numbers, in this context, are called *scalars*). If the scalar is negative, the vector's direction is flipped as a result of the multiplication. Since Δt in the definition of velocity is always positive, it follows that the average velocity vector always points in the same direction as the displacement, which makes sense.

To get the instantaneous velocity, you just take the limit of the expression (1.2.13) as $\Delta t \rightarrow 0$, for each component separately. The resulting vector $\vec{v}$ has components $v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}$, etc., which can also be written as $v_x = dx/dt$, $v_y = dy/dt$, . . .

All the results derived above hold for each spatial dimension and its corresponding velocity component. For instance, the graphical method shown in Figure 1.2.5 can always be used to get Δx if the function $v_x(t)$ is known, or equivalently to get Δy if you know $v_y(t)$, and so on.

Introducing the velocity vector at this point does cause a little bit of a notational difficulty. For quantities like x and Δx , it is pretty obvious that they are the x components of the vectors $\vec{r}$ and $\Delta \vec{r}$ respectively; however, the quantity that we have so far been calling simply v should more properly be denoted as v_x (or v_y if the motion is along the y axis). In fact, there is a convention that if you use the symbol for a vector without the arrow on top or any x , y , . . . subscripts, you must mean the *magnitude* of the vector. In this book, however, I have decided *not* to follow that convention, at least not until we get to [Chapter 8](#) (and even then I will use it only for forces). This is because we will spend most of our time dealing with motion in only one dimension, and it makes the notation unnecessarily cumbersome to keep having to write the x or y subscripts on every component of every vector, when you really only have one dimension to worry about in the first place. So v will, throughout, refer to the relevant component of the velocity vector, to be inferred from the context, until we get to [Chapter 8](#) and actually need to deal with both a v_x and a v_y explicitly.

Finally, notice that the magnitude of the velocity vector, $|\vec{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2}$, is equal to the *instantaneous speed*, since, as $\Delta t \rightarrow 0$, the magnitude of the displacement vector, $|\Delta \vec{r}|$, becomes the actual distance traveled by the object in the time interval Δt .

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1.3: Reference Frame Changes and Relative Motion

Everything up to this point assumes that we are using a fixed, previously agreed upon **reference frame**. Basically, this is just an origin and a set of axes along which to measure our coordinates, as shown in [Figure 1.2.1](#).

There are, however, a number of situations in physics that call for the use of different reference frames, and, more importantly, that require us to *convert* various physical quantities from one reference frame to another. For instance, imagine you are on a boat on a river, rowing downstream. You are moving with a certain velocity relative to the water around you, but the water itself is flowing with a different velocity relative to the shore, and your actual velocity relative to the shore is the sum of those two quantities. Ships generally have to do this kind of calculation all the time, as do airplanes: the “airspeed” is the speed of a plane relative to the air around it, but that air is usually moving at a substantial speed relative to the earth.

The way we deal with all these situations is by introducing two reference frames, which here I am going to call A and B. One of them, say A, is “at rest” relative to the earth, and the other one is “at rest” relative to something else—which means, really, moving along with that something else. (For instance, a reference frame at rest “relative to the river” would be a frame that’s moving along with the river water, like a piece of driftwood that you could measure your progress relative to.)

In any case, graphically, this will look as in [Figure 1.3.1](#), which I have drawn for the two-dimensional case because I think it makes it easier to visualize what’s going on:

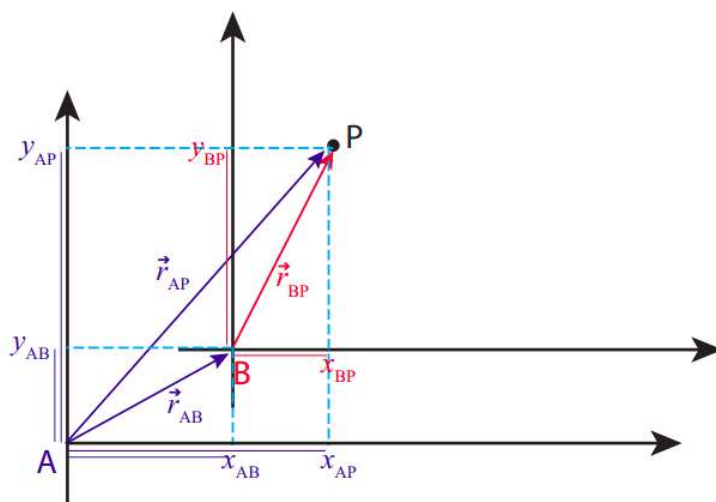


Figure 1.3.1: Position vectors and coordinates of a point P in two different reference frames, A and B.

In the reference frame A, the point P has position coordinates (x_{AP}, y_{AP}) . Likewise, in the reference frame B, its coordinates are (x_{BP}, y_{BP}) . As you can see, the notation chosen is such that every coordinate in A will have an “A” as a first subscript, while the second subscript indicates the object to which it refers, and similarly for coordinates in B.

The coordinates (x_{AB}, y_{AB}) are special: they are the coordinates, in the reference frame A, of the origin of reference frame B. This is enough to fully locate the frame B in A, as long as the frames are not rotated relative to each other.

The thin colored lines I have drawn along the axes in [Figure 1.3.1](#) are intended to make it clear that the following equations hold:

$$\begin{aligned}x_{AP} &= x_{AB} + x_{BP} \\y_{AP} &= y_{AB} + y_{BP}\end{aligned}\tag{1.3.1}$$

Although the figure is drawn for the easy case where all these quantities are positive, you should be able to convince yourself that Eqs. (1.3.1) hold also when one or more of the coordinates have negative values.

All these coordinates are also the components of the respective position vectors, shown in the figure and color-coded by reference frame (so, for instance, $\vec{r}_{AP}$ is the position vector of P in the frame A), so the equations (1.3.1) can be written more compactly as the single vector equation

$$\vec{r}_{AP} = \vec{r}_{AB} + \vec{r}_{BP}.\tag{1.3.2}$$

From all this you can see how to add vectors: algebraically, you just add their components separately, as in Eqs. (1.3.1); graphically, you draw them so the tip of one vector coincides with the tail of the other (we call this “tip-to-tail”), and then draw the sum vector from the tail of the first one to the tip of the other one. (In general, to get two arbitrary vectors tip-to-tail you may need to displace one of them; this is OK provided you do not change its orientation, that is, provided you only displace it, not rotate it. We’ll see how this works in a moment with velocities, and later on with forces.)

Of course, I showed you already how to *subtract* vectors with Figure 1.2.3: again, algebraically, you just subtract the corresponding coordinates, whereas graphically you draw them with a common origin, and then draw the vector from the tip of the vector you are subtracting to the tip of the other one. If you read the previous paragraph again, you can see that Figure 1.2.3 can equally well be used to show that $\Delta\vec{r} = \vec{r}_f - \vec{r}_i$, as to show that $\vec{r}_f = \vec{r}_i + \Delta\vec{r}$.

In a similar way, you can see graphically from Figure 1.3.1 (or algebraically from Equation (1.3.2)) that the position vector of P in the frame B is given by $\vec{r}_{BP} = \vec{r}_{AP} - \vec{r}_{AB}$. The last term in this expression can be written in a different way, as follows. If I follow the convention I have introduced above, the quantity x_{BA} (with the order of the subscripts reversed) would be the x coordinate of the origin of frame A in frame B, and algebraically that would be equal to $-x_{AB}$, and similarly $y_{BA} = -y_{AB}$. Hence the vector equality $\vec{r}_{AB} = -\vec{r}_{BA}$ holds. Then,

$$\vec{r}_{BP} = \vec{r}_{AP} - \vec{r}_{AB} = \vec{r}_{AP} + \vec{r}_{BA}. \quad (1.3.3)$$

This is, in a way, the “inverse” of Equation (1.3.2): it tells us how to get the position of P in the frame B if we know its position in the frame A.

Let me show next you how all this extends to displacements and velocities. Suppose the point P indicates the position of a particle at the time t . Over a time interval Δt , both the position of the particle and the relative position of the two reference frames may change. We can add yet another subscript, i or f , (for initial and final) to the coordinates, and write, for example,

$$\begin{aligned} x_{AP,i} &= x_{AB,i} + x_{BP,i} \\ x_{AP,f} &= x_{AB,f} + x_{BP,f} \end{aligned} \quad (1.3.4)$$

Subtracting these equations gives us the corresponding displacements:

$$\Delta x_{AP} = \Delta x_{AB} + \Delta x_{BP}. \quad (1.3.5)$$

Dividing Equation (1.3.5) by Δt we get the average velocities¹, and then taking the limit $\Delta t \rightarrow 0$ we get the instantaneous velocities. This applies in the same way to the y coordinates, and the result is the vector equation

$$\vec{v}_{AP} = \vec{v}_{BP} + \vec{v}_{AB}. \quad (1.3.6)$$

I have rearranged the terms on the right-hand side to (hopefully) make it easier to visualize what’s going on. In words: the velocity of the particle P relative to (or *measured in*) frame A is equal to the (vector) sum of the velocity of the particle as measured in frame B, plus the velocity of frame B relative to frame A.

The result (1.3.6) is just what we would have expected from the examples I mentioned at the beginning of this section, like rowing in a river or an airplane flying in the wind. For instance, for the airplane $\vec{v}_{BP}$ could be its “airspeed” (only it has to be a vector, so it would be more like its “airvelocity”: that is, its velocity relative to the air around it), and $\vec{v}_{AB}$ would be the velocity of the air relative to the earth (the wind velocity, at the location of the airplane). In other words, A represents the earth frame of reference and B the air, or wind, frame of reference. Then, $\vec{v}_{AP}$ would be the “true” velocity of the airplane relative to the earth. You can see how it would be important to add these quantities as vectors, in general, by considering what happens when you fly in a cross wind, or try to row across a river, as in Figure 1.3.2 below.

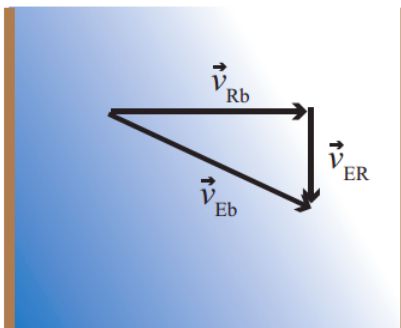


Figure 1.3.2: Rowing across a river. If you head “straight across” the river (with velocity vector $\vec{v}_{Rb}$ in the moving frame of the river, which is flowing with velocity $\vec{v}_{ER}$ in the frame of the earth), your actual velocity relative to the shore will be the vector $\vec{v}_{Eb}$. This is an instance of Equation (1.3.6), with frame A being E (the earth), frame B being R (the river), and “b” (for “boat”) standing for the point P we are tracking.

As you can see from this couple of examples, Equation (1.3.6) is often useful as it is written, but sometimes the information we have is given to us in a different way: for instance, we could be given the velocity of the object in frame A ($\vec{v}_{AP}$), and the velocity of frame B as seen in frame A ($\vec{v}_{AB}$), and told to calculate the velocity of the object as seen in frame B. This can be easily accomplished if we note that the vector $\vec{v}_{AB}$ is equal to $-\vec{v}_{BA}$; that is to say, the velocity of frame B as seen from frame A is just the opposite of the velocity of frame A as seen from frame B. Hence, Equation (1.3.6) can be rewritten as

$$\vec{v}_{AP} = \vec{v}_{BP} - \vec{v}_{BA}. \quad (1.3.7)$$

For most of the next few chapters we are going to be considering only motion in one dimension, and so we will write Equation (1.3.6) (or (1.3.7)) without the vector symbols, and it will be understood that v refers to the component of the vector $\vec{v}$ along the coordinate axis of interest.

A quantity that will be particularly important later on is the *relative velocity* of two objects, which we could label 1 and 2. The velocity of object 2 relative to object 1 is, by definition, the velocity which an observer moving along with 1 would measure for object 2. So it is just a simple frame change: let the earth frame be frame E and the frame moving with object 1 be frame 1, then the velocity we want is v_{12} (“velocity of object 2 in frame 1”). If we make the change $A \rightarrow 1$, $B \rightarrow E$, and $P \rightarrow 2$ in Equation (1.3.7), we get

$$v_{12} = v_{E2} - v_{E1}. \quad (1.3.8)$$

In other words, the velocity of 2 relative to 1 is just the velocity of 2 minus the velocity of 1. This is again a familiar effect: if you are driving down the highway at 50 miles per hour, and the car in front of you is driving at 55, then its velocity relative to you is 5 mph, which is the rate at which it is moving away from you (in the forward direction, assumed to be the positive one).

It is important to realize that all these velocities are *real* velocities, each in its own reference frame. Something may be said to be truly moving at some velocity in one reference frame, and just as truly moving with a different velocity in a different reference frame. I will have a lot more to say about this in the next chapter, but in the meantime you can reflect on the fact that, if a car moving at 55 mph collides with another one moving at 50 mph in the same direction, the damage will be basically the same as if the first car had been moving at 5 mph and the second one had been at rest. For practical purposes, where you are concerned, another car’s velocity relative to yours is that car’s “real” velocity.

Resources

A good app for practicing how to add vectors (and how to break them up into components, magnitude and direction, etc.) may be found here: <https://phet.colorado.edu/en/simulation/vector-addition>.

Perhaps the most dramatic demonstration of how Equation (1.3.6) works in the real world is in this episode of *Mythbusters*: <https://www.youtube.com/watch?v=BLuI118nhzc>. (If this link does not work, do a search for “Mythbusters cancel momentum.”) They shoot a ball from the bed of a truck, with a velocity (relative to the truck) of 60 mph backwards, while the truck is moving forward at 60 mph. I think the result is worth watching. (Do not be distracted by their talk about momentum. We will get there, in time.)

A very old, but also very good, educational video about different frames of reference is this one: <https://www.youtube.com/watch?v=sS17fCom0Ns>. You should try to watch at least part of it. Many things will be relevant to later parts of the course, including projectile motion, and the whole discussion of relative motion coming up next, in [Chapter 2](#).

¹ We have made a very natural assumption, that the time interval Δt is the same for observers tracking the particle's motion in frames A and B, respectively (where each observer is understood to be moving along with his or her frame). This, however, turns out to be *not* true when any of the velocities involved is close to the speed of light, and so the simple addition of velocities formula (1.3.6) does not hold in Einstein's relativity theory. (This is actually the first bit of real physics I have told you about in this book, so far; unfortunately, you will have no use for it this semester!)

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1.4: In Summary

1. To describe the motion of an object in one dimension we treat it as a mathematical point, and consider its *position coordinate*, x (often shortened to just the *position*), as a function of time: $x(t)$.
2. Numerically, the position coordinate is the distance to a chosen origin, with a positive or negative sign depending on which side of the origin the point is. For every problem, when we introduce a coordinate axis we need to specify a positive direction. Starting from the origin in that direction, the position coordinate is positive and increasing, whereas going from the origin in the opposite direction (negative direction) it becomes increasingly negative.
3. The displacement of an object over a time interval from an initial time t_i to a final time t_f is the quantity $\Delta x = x_f - x_i$, where x_f is the position of the object at the final time (or, the final position), and x_i the position at the initial time (or initial position).
4. The average velocity of an object over the time interval from t_i to t_f is defined as $v_{av} = \Delta x / \Delta t$, where $\Delta t = t_f - t_i$.
5. The *instantaneous velocity* (often just called the *velocity*) of an object at the time t is the limit value of the quantity $\Delta x / \Delta t$, calculated for successively shorter time intervals Δt , all with the same initial time $t_i = t$. This is, mathematically, the definition of the derivative of the function $x(t)$ at the time t , which we express as $v = dx/dt$.
6. Graphically, the instantaneous velocity of the object at the time t is the slope of the tangent line to the x -vs- t graph at the time t .
7. The instantaneous velocity of an object is a positive or negative quantity depending on whether the object, at that instant, is moving in the positive or the negative direction.
8. For an object moving with constant velocity v , the position function is given by [Equation (1.2.10)]:

$$x(t) = x_i + v(t - t_i)$$

where t_i is an arbitrarily chosen initial time and x_i the position at that time. This can also be written in the form given by Equation (1.2.9). The argument (t) on the left-hand side of Equation (1.2.10) is optional, and t_i is often set equal to zero, giving just $x = x_i + vt$. This, however, is not quite as generally applicable as the result Equation (1.2.9) or Equation (1.2.10).

9. For an object moving with changing velocity, the total displacement in between times t_i and t_f is equal to the total area under the v -vs- t curve in between those times; areas below the horizontal (t) axis must be treated as negative.
10. In two or more dimensions one introduces, for every point in space, a *position vector* whose components are just the Cartesian coordinates of that point; then the displacement vector is defined as $\Delta \vec{r} = \vec{r}_f - \vec{r}_i$, the average velocity vector is $\vec{v}_{av} = \Delta \vec{r} / \Delta t$, and the instantaneous velocity vector is the limit of this as Δt goes to zero. Vectors are added by adding their components separately; to multiply a vector by an ordinary number, or *scalar*, we just multiply each component by that number.
11. When tracking the motion of an object, "P", in two different reference frames, A and B, the position vectors are related by $\vec{r}_{AP} = \vec{r}_{AB} + \vec{r}_{BP}$, and likewise the velocity vectors: $\vec{v}_{AP} = \vec{v}_{AB} + \vec{v}_{BP}$. Here, the first subscript tells you in which reference frame you are measuring, and the second subscript what it is that you are looking at; $\vec{r}_{AB}$ is the position vector of the origin of frame B as seen in frame A, and $\vec{v}_{AB}$ its velocity.

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1.5: Examples

Example 1.5.1: Motion with (Piecewise) Constant Velocity

You leave your house on your bicycle to go visit a friend. At your normal speed of 9 mph, you know it takes you 6 minutes to get there. This time, though, when you have traveled half the distance you realize you forgot a book at home that you were going to return to your friend, so you turn around and pedal at twice your normal speed, get back home, grab the book, and start off again for your friend's house at 18 mph (imagine you are really fit to pull this off!)

- How far away from you does your friend live?
- What is the total distance you travel on this trip?
- How long did the whole trip take?
- Draw a position versus time and a velocity versus time graph for the whole trip. Use SI units for both graphs. Neglect the time it takes you to stop and turn around, and also the time it takes you to run into your house and grab the book (in other words, assume those changes in your direction of motion happen instantly).
- Show explicitly, using your v -vs- t graph, that the graphical method of [Figure 1.2.5](#) gives you the total displacement for your trip.

Solution

I am going to work out this problem using both miles and SI units, the first because it seems most natural, and the second because we are asked to use SI units for part (d), so we might as well use them from the start. In general, you should use SI units whenever you can. If you are unsure of what to do in a specific problem, ask your instructor!

(a) We are told that at 9 miles per hour it would take 6 minutes to get there, so let us use

$$\Delta x = v\Delta t \quad (1.5.1)$$

with $v = 9$ mph and $\Delta t = 6$ min. We have to either convert the hours to minutes, or vice-versa. Again, in this case it seems easiest to realize that 6 min equals 1/10 of an hour, so:

$$\Delta x = \left(9 \frac{\text{miles}}{\text{hr}}\right) \times 0.1 \text{ hr} = 0.9 \text{ miles.} \quad (1.5.2)$$

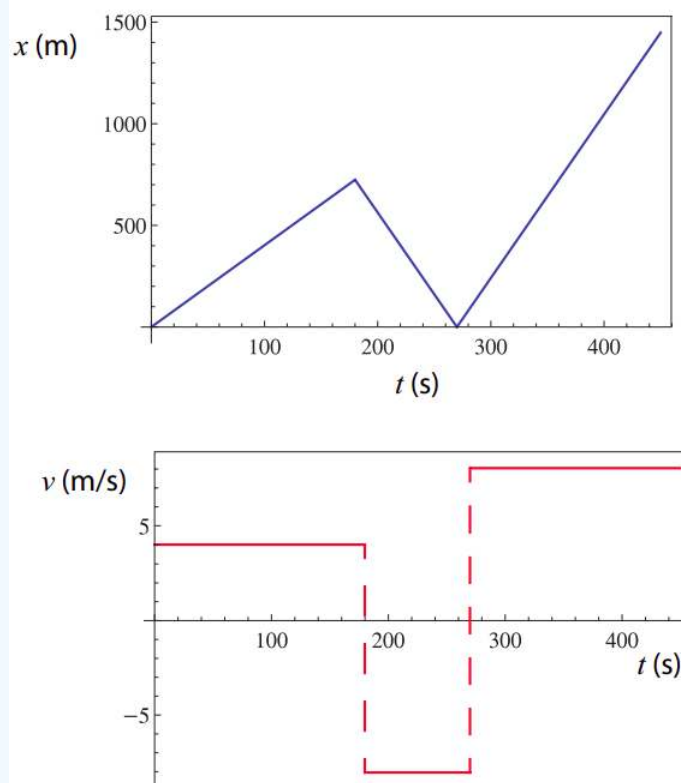
In SI units, 9 mph = 4.023 m/s, and 6 min = 360 s, so $\Delta x = 1448$ m.

(b) This is just a matter of keeping track of the distance traveled in the various parts of the trip. You start by riding half the distance to your friend's house, which is to say, 0.45 miles, and then you ride that again back home, so that's 0.9 miles, and then you're back where you started, so you still have to go the 0.9 miles to your friend's house. So overall, you ride for 1.8 miles, or 2897 m.

(c) The whole trip consists, as detailed above, of 0.45 miles at 9 mph, and the rest, which is 1.35 miles, at 18 mph. Applying $\Delta t = \Delta x/v$ to each of these intervals, we get a total time of

$$\begin{aligned} \Delta t &= \frac{0.45 \text{ miles}}{9 \text{ mph}} + \frac{1.35 \text{ miles}}{18 \text{ mph}} = 0.125 \text{ hours} \\ &= 0.125 \times 60 \text{ min} = 7.5 \text{ min} \\ &= 7.5 \times 60 \text{ s} = 450 \text{ s} \end{aligned} \quad (1.5.3)$$

(d) The graphs are shown below. Details on how to get them follow.



- First interval: from $t = 0$ to $t = 180$ s (3 min, which is what it would take to cover half the distance to your friend's house at 9 mph). The velocity is a constant $v = 4.023$ m/s. For the position graph, use Equation (1.2.10) with $x_i = 0$, $t_i = 0$ and $v = 4.023$ m/s.
- Second interval: from $t = 180$ s to $t = 270$ s (it takes you half of 3 min, which is to say 90 s, to cover the same distance as above at twice the speed). The velocity is a constant $v = -8.046$ m/s (twice what it was earlier, but in the opposite direction). For the position graph, use Equation (1.2.10) with $x_i = 724$ m (this is half of the distance to your friend's house, and the starting position for this interval), $t_i = 180$ s and $v = -8.046$ m/s.
- Third interval: from $t = 270$ s to $t = 450$ s. The velocity is a constant $v = 8.046$ m/s (same speed as just before, but in the opposite direction). For the position graph, use Equation (1.2.10) with $x_i = 0$ m (you start back at your house), $t_i = 270$ s and $v = 8.046$ m/s.

If you are familiar with the software package *Mathematica*, the position graph was produced using the command

```
Plot[If[t<180, 4.023 t, If[t<270, 4.023*180-8.046 (t-180), 8.046 (t-270)]], {t, 0, 450}]
```

and the velocity graph was produced using

```
Plot[If[t<180, 4.023, If[t<270, -8.046, 8.046]], {t, 0, 450}]
```

(and then connecting the horizontal lines by hand, which is not necessary, but helps to visualize what's going on).

The graphs could also have been produced using the free plotting software package Gnuplot (available here: <http://www.gnuplot.info/download.html>) with the following commands:

```
gnuplot> set dummy t
gnuplot> f(t) = t<180 ? 4.023*t : t<270 ? 4.023*180-8.046*(t-180) : 8.046*(t-270)
gnuplot> plot [0:450] f(t)
```

The first line sets the default independent variable to t (instead of x , which is what Gnuplot expects). The second line defines the piecewise function using the *ternary operator* ($? :$) borrowed from the C programming language. The third line plots the function over the range indicated.

(e) For this we need to find the area under the v -vs- t graph we just plotted. Basically, we have three rectangles: the first one has base 180 units (s) and height 4 units (m/s), so its area is $4 \times 180 = 720$ (m). The second rectangle has base 90 units and height -8 (negative, because it is below the horizontal axis!), so its area is -720 . The last one has base 180 units again (from 270 to 450) and height 8, so its area is $8 \times 180 = 1440$. So the total area “under” the v -vs- t curve is

$$720 - 720 + 1440 = 1440 \text{ meters}$$

which is (approximately) your total displacement, that is, the 9 miles to your friend’s house. (Of course, we would have obtained a more accurate result if we had used the more accurate values for the “heights” of 4.023, -8.046 , and 8.046, but if all we have to go by is the graph, such accuracy is pretty much impossible.)

Example 1.5.2: Addition of velocities, relative motion

This example was inspired by the “race on a moving sidewalk” demo at <http://physics.bu.edu/~duffy/classroom.html>. Please go take a look at it!

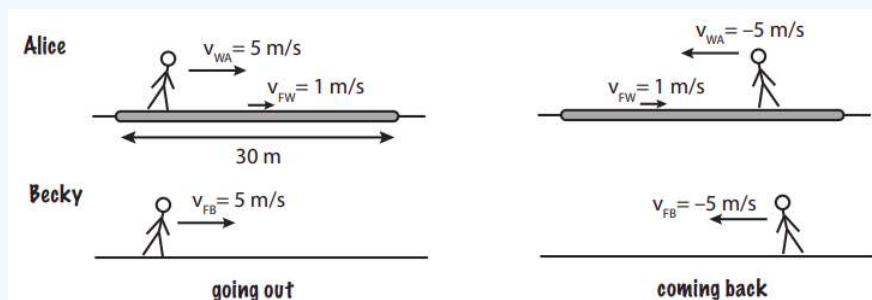
Two girls, Ann and Becky (yes, A and B) decide to have a race while they wait for a plane at a nearly-deserted airport. Ann will run on the moving walkway, to the end of it (which is 30 m away) and back, whereas Becky will run alongside her on the (non-moving) floor, also 30 m out and back. The walkway moves at 1 m/s, and the girls both run at the same constant speed of 5 m/s *relative to the surface they are standing on*.

- Relative to the (non-moving) floor, what is Ann’s velocity for the first leg of her race, when she is moving in the same direction as the walkway (take that to be the positive direction)? What is her velocity for the return leg?
- How long does it take each of the girls to complete their race?
- When both girls are running in the positive direction, what is Becky’s velocity relative to Ann? (That is, how fast does Ann see Becky move, and in what direction?)
- When Ann turns around and starts running in the negative direction, but Becky is still running in the positive direction, what is Becky’s velocity relative to Ann?
- What is the total distance Ann runs *in the moving walkway’s frame of reference*?

Solution

I am going to solve this in the format that you will be required to use this semester for most of the homework and exam problems. I will not be able to do this for every single example, but you should! Please follow this carefully.

To begin with, you must draw a sketch of the situation described in the problem, detailed enough to include all the relevant information you are given. Here is mine:



Note that I have drawn one picture for each half of the race, and that all the information given in the text of the problem is there. The figure makes it clear also the notation I will be using for each of the girls’ velocities, and to see at a glance what is happening.

You should next state what kind of problem this is and what basic result (theorem, principle, or equation(s)) you are going to use to solve it. For this problem, you could say:

“This is a relative motion/reference frame transformation problem. I will use Equation (1.3.6)

$$\vec{v}_{AP} = \vec{v}_{BP} + \vec{v}_{AB}$$

as well as the basic equation for motion with constant velocity:”

$$\Delta x = v \Delta t.$$

After that, solve each part in turn, and make sure to show all your work!

(a) Let F stand for the floor frame of reference, and W the walkway frame. In the notation of [Section 1.3](#), we have $v_{FW} = 1 \text{ m/s}$. For the first leg of her race, we are told that Ann's velocity *relative to the walkway* is 5 m/s , so $v_{WA} = 5 \text{ m/s}$. Then, by Equation (1.3.6) (with the following change of indices: $A \rightarrow F$, $B \rightarrow W$, and $P \rightarrow A$),

$$v_{FA} = v_{FW} + v_{WA} = 1 \frac{\text{m}}{\text{s}} + 5 \frac{\text{m}}{\text{s}} = 6 \frac{\text{m}}{\text{s}} \quad (1.5.4)$$

(when you see an equation like this, full of subscripts, it is a good practice to read it out, mentally, to yourself: “Ann’s velocity relative to the floor equals the velocity of the walkway relative to the floor plus Ann’s velocity relative to the walkway.” Then take a moment to see if it makes sense! Here is a place where the picture can be really helpful.)

For the return leg, use the same formula, but note that now her velocity relative to the walkway is *negative*, $v_{WA} = -5 \text{ m/s}$, since she is moving in the opposite direction:

$$v_{FA} = v_{FW} + v_{WA} = 1 \frac{\text{m}}{\text{s}} - 5 \frac{\text{m}}{\text{s}} = -4 \frac{\text{m}}{\text{s}}. \quad (1.5.5)$$

(b) Relative to the floor reference frame, we have just seen that Ann first covers 30 m at a speed of 6 m/s , and then the same 30 m at a speed of 4 m/s , so her total time is

$$\Delta t_A = \frac{30 \text{ m}}{6 \text{ m/s}} + \frac{30 \text{ m}}{4 \text{ m/s}} = 5 \text{ s} + 7.5 \text{ s} = 12.5 \text{ s} \quad (1.5.6)$$

whereas Becky just runs 30 m at 5 m/s both ways, so it takes her 6 s either way, for a total of 12 s , which means she wins the race.

(c) The quantity we want is written, in the notation of [Section 1.3](#), v_{AB} (“velocity of Becky relative to Ann”). To calculate this, we just need to know the velocities of both girls in some frame of reference (the same for both!), then subtract Ann’s velocity from Becky’s (this is what Equation (1.3.8) is saying). In this case, if we just choose the floor’s reference frame, we have $v_{FA} = 6 \text{ m/s}$ and $v_{FB} = 5 \text{ m/s}$, so

$$v_{AB} = v_{FB} - v_{FA} = 5 \frac{\text{m}}{\text{s}} - 6 \frac{\text{m}}{\text{s}} = -1 \frac{\text{m}}{\text{s}} \quad (1.5.7)$$

The negative sign makes sense: Ann sees Becky *falling behind her*, so relative to her Becky is moving *backwards*, which is to say, in the direction we have identified as negative.

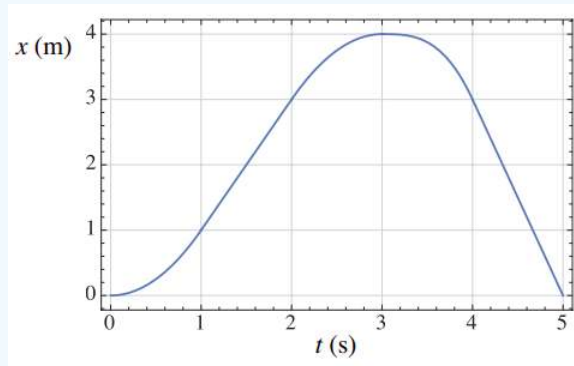
(d) Again we use the same equation, and Becky’s velocity is still the same, but now Ann’s velocity is $v_{FA} = -4 \text{ m/s}$ (note the negative sign!), so

$$v_{AB} = v_{FB} - v_{FA} = 5 \frac{\text{m}}{\text{s}} - \left(-4 \frac{\text{m}}{\text{s}}\right) = 9 \frac{\text{m}}{\text{s}} \quad (1.5.8)$$

(e) You may find this a bit surprising, but if you think about it the explanation for why Ann lost the race, despite her running at the same speed as Becky *relative to the surface she was standing on*, has to be that she actually ran a longer distance on that surface! Since she was running for a total of 12.5 s at a constant *speed* (not velocity!) of 5 m/s in the walkway frame, then in that frame she ran a distance $d = |v|\Delta t = 5 \times 12.5 = 62.5 \text{ m}$. That is the total length of walkway that she actually stepped on.

1.6: Exercises

Exercise 1.6.1



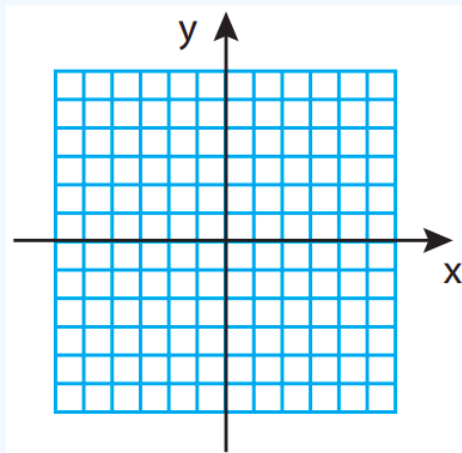
The above figure is the position (in meters) versus time (in seconds) graph of an object in motion. Only the segments between $t = 1$ s and $t = 2$ s, and between $t = 4$ s and $t = 5$ s, are straight lines. The peak of the curve is at $t = 3$ s, $x = 4$ m.

Answer the following questions, and provide a brief justification for your answer in every case.

- At what time(s) is the object's velocity equal to zero?
- For what range(s) of times is the object moving with constant velocity?
- What is the object's position coordinate at $t = 1$ s?
- What is the displacement of the object between $t = 1$ s and $t = 4$ s?
- What is the distance traveled between $t = 1$ s and $t = 4$ s?
- What is the instantaneous velocity of the object at $t = 1.5$ s?
- What is its average velocity between $t = 1$ s and $t = 3$ s?

Exercise 1.6.2

A particle is initially at $x_i = 3$ m, $y_i = -5$ m, and after a while it is found at the coordinates $x_f = -4$ m, $y_f = 2$ m.



- On the grid above, draw the initial and final position vectors, and the displacement vector.
- What are the components of the displacement vector?
- What are the magnitude and direction of the displacement vector? (You can specify the direction by the angle it makes with either the positive x or the positive y axis.)

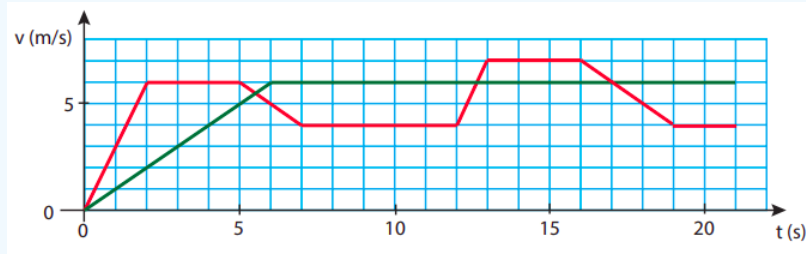
Exercise 1.6.3

Marshall Dillon is riding at 30 mph after the robber of the Dodge City bank, who has a head start of 15 minutes, but whose horse can only make 25 mph on a good day. How long does it take for Dillon to catch up with the bad guy, and how far from Dodge City are they when this happens? (Assume the road is straight, for simplicity.)

Exercise 1.6.4

The picture below shows the velocity versus time graph of the first 21 seconds of a race between two friends, “Red” and “Green.”

- Who is ahead at $t = 10$ s, and by how much?
- Who passes the 100 m marker first?



Exercise 1.6.5

You are trying to pass a truck on the highway. The truck is driving at 55 mph, so you speed up to 60 mph and move over to the left lane. If the truck is 17 m long, and your car is 3 m long

- how long does it take you to pass the truck completely?
- How far (along the highway) have you traveled in that time? Note: to answer part (a) look at the problem from the perspective of the truck driver. How far are you going relative to him, and how far would it take you to cover 20 m at that speed?

Exercise 1.6.6

Suppose the position function of a particle moving in one dimension is given by

$$x(t) = 5 + 3t + 2t^2 - 0.5t^3 \quad (1.6.1)$$

where the coefficients are such that the result will be in meters if you enter the time in seconds. What is the particle's velocity at $t = 2$ s? There are two ways you can do this:

- If you know calculus, calculate the derivative of Eq (1.6.1) and evaluate it at $t = 2$ s.
- If you do *not* yet know how to take derivatives, calculate the limit in the definition (1.2.8). That is to say, calculate $\Delta x / \Delta t$ with $t_i = 2$ s and Δt equal, first, to 0.1 s, then to 0.01 s, and then to 0.001 s. You will need to keep more than the usual 4 decimals in the intermediate calculations if you want an accurate result, but you should still report only 3 significant digits in the final result.

Exercise 1.6.7

Suppose you are rowing across a river, as in Figure 1.3.2. Your speed is 2 miles per hour relative to the current, which is moving at a leisurely 1 mile per hour. If the river is 10 m wide,

- How far downstream do you end up?
- To row straight across you would need to have an upstream velocity component (relative to the current). How large would that be?
- If your rowing speed is still only 2 miles per hour, how long does it take you to row across the river now?

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CHAPTER OVERVIEW

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2.1: The Law of Inertia

There is something funny about motion with constant velocity: it is *indistinguishable from rest*. Of course, you can usually tell whether you are moving *relative to something else*. But if you are enjoying a smooth airplane ride, without looking out the window, you have no idea how fast you are moving, or even, indeed (if the flight is exceptionally smooth) whether you are moving at all. I am actually writing this on an airplane. The flight screen informs me that I am moving at 480 mph relative to the ground, but I do not feel anything like that: just a gentle rocking up and down and sideways that gives me no clue as to what my forward velocity is.

If I were to drop something, I know from experience that it would fall on a straight line—relative to me, that is. If it falls from my hand it will land at my feet, just as if we were all at rest. But we are *not* at rest. In the half second or so it takes for the object to fall, the airplane has moved forward 111 meters relative to the ground. Yet the (hypothetical) object I drop does not land 300 feet behind me! It moves forward with me as it falls, even though I am not touching it. *It keeps its initial forward velocity*, even though it is no longer in contact with me or anything connected to the airplane.

(At this point you might think that the object is still in contact with the air inside the plane, which is moving with the plane, and conjecture that maybe it is the air inside the plane that “pushes forward” on the object as it falls and keeps it from moving backwards. This is not necessarily a dumb idea, but a moment’s reflection will convince you that it is impossible. We are all familiar with the way air pushes on things moving through it, and we know that the force an object experiences depends on its mass and its shape, so if that was what was happening, dropping objects of different masses and shapes I would see them falling in all kind of different ways—as I would, in fact, if I were dropping things from rest outdoors in a strong wind. But that is not what we experience on an airplane at all. The air, in fact, has no effect on the forward motion of the falling object. It does not push it in any way, because it is moving at the same velocity. This, in fact, reinforces our previous conclusion: the object keeps its forward velocity while it is falling, in the absence of any external influence.)

This remarkable observation is one of the most fundamental principles of physics (yes, we have started to learn physics now!), which we call the **law of inertia**. It can be stated as follows: in the absence of any external influence (or *force*) acting on it, an object at rest will stay at rest, while an object that is already moving with some velocity will keep that same velocity (speed and direction of motion)—at least until it is, in fact, acted upon by some force.

Please let that sink in for a moment, before we start backtracking, which we have to do now on several accounts. First, I have used repeatedly the term “force,” but I have not defined it properly. Or have I? What if I just said that forces are precisely any “external influences” that may cause a change in the velocity of an object? That will work, I think, until it is time to explore the concept in more detail, a few chapters from now.

Next, I need to draw your attention to the fact that the object I (hypothetically) dropped did not actually keep its *total* initial velocity: it only kept its initial *forward* velocity. In the downward direction, it was speeding up from the moment it left my hand, as would any other falling object (and as we shall see later in this chapter). But this actually makes sense in a certain way: there was no forward force, so the forward velocity remained constant; there was, however, a vertical force acting all along (the force of gravity), and so the object did speed up in that direction. This observation is, in fact, telling us something profound about the world’s geometry: namely, that forces and velocities are *vectors*, and laws such as the law of inertia will typically apply to the vector as a whole, as well as to each component separately (that is to say, each dimension of space). This anticipates, in fact, the way we will deal, later on, with motion in two or more dimensions; but we do not need to worry about that for a few chapters still.

Finally, it is worth spending a moment reflecting on how radically the law of inertia seems to contradict our intuition about the way the world works. What it seems to be telling us is that, if we throw or push an object, it should continue to move forever with the same speed and in the same direction with which it set out—something that we know is certainly not true. But what’s happening in “real life” is that, just because we have left something alone, it doesn’t mean the world has left it alone. After we lose contact with the object, all sorts of other forces will continue to act on it. A ball we throw, for instance, will experience air resistance or drag (the same effect I was worrying about in that paragraph in parenthesis in the previous page), and that will slow it down. An object sliding on a surface will experience friction, and that will slow it down too. Perhaps the closest thing to the law of inertia in action that you may get to see is a hockey puck sliding on the ice: it is remarkable (perhaps even a bit frightening) to see how little it slows down, but even so the ice does exert a (very small) frictional force that would bring the puck to a stop eventually.

This is why, historically, the law of inertia was not discovered until people started developing an appreciation for frictional forces, and the way they are constantly acting all around us to oppose the relative motion of any objects trying to slide past each other.

This mention of relative motion, in a way, brings us full circle. Yes, relative motion is certainly detectable, and for objects in contact it actually results in the occurrence of forces of the frictional, or drag, variety. But *absolute* (that is, without reference to anything external) motion with constant velocity is fundamentally undetectable. And in view of the law of inertia, it makes sense: if no force is required to keep me moving with constant velocity, it follows that as long as I am moving with constant velocity I should not be feeling any net force acting on me; nor would any other detection apparatus I might be carrying with me.

What we do feel in our bodies, and what we can detect with our inertial navigation systems (now you may start to guess why they are called “inertial”), is a *change* in our velocity, which is to say, our *acceleration* (to be defined properly in a moment). We rely, ultimately, on the law of inertia to detect accelerations: if my plane is shaking up and down, because of turbulence (as, in fact, it is right now!), the water in my cup may not stay put. Or, rather, the water may try to stay put (really, to keep moving, at any moment, with whatever velocity it has at that moment), but if the cup, which is connected to my hand which is connected, ultimately, to this bouncy plane, moves suddenly out from under it, not all of the water’s parts will be able to adjust their velocities to the new velocity of the cup in time to prevent a spill.

This is the next very interesting fact about the physical world that we are about to discover: forces cause accelerations, or changes in velocity, but they do so in different degrees for different objects; and, moreover, the ultimate change in velocity *takes time*. The first part of this statement has to do with the concept of *inertial mass*, to be introduced in the next chapter; the second part we are going to explore right now, after a brief detour to define *inertial reference frames*.

Inertial Reference Frames

The example I just gave you of what happens when a plane in flight experiences turbulence points to an important phenomenon, namely, that there may be times where the law of inertia may not *seem* to apply in a certain reference frame. By this I mean that an object that I left at rest, like the water in my cup, may suddenly start to move—relative to the reference frame coordinates—even though nothing and nobody is acting on it. More dramatically still, if a car comes to a sudden stop, the passengers may be “projected forward”—they were initially at rest relative to the car frame, but now they find themselves moving forward (always in the car reference frame), to the point that, if they are not wearing seat belts, they may end up hitting the dashboard, or the seat in front of them.

Again, nobody has pushed on them, and in fact what we can see in this case, from outside the car, is nothing but the law of inertia at work: the passengers were just keeping their initial velocity, when the car suddenly slowed down under and around them. So there is nothing wrong with the law of inertia, but *there is a problem with the reference frame*: if I want to describe the motion of objects in a reference frame like a plane being shaken up or a car that is speeding up or slowing down, I need to allow for the fact that objects may move—always relative to that frame—in an *apparent* violation of the law of inertia.

The way we deal with this in physics is by introducing the very important concept of an *inertial reference frame*, by which we mean a reference frame in which all objects will, at all times, be observed to move (or not move) in a way fully consistent with the law of inertia. In other words, the law of inertia has to hold *when we use that frame’s own coordinates to calculate the objects’ velocities*. This, of course, is what we always do instinctively: when I am on a plane I locate the various objects around me relative to the plane frame itself, not relative to the distant ground.

To ascertain whether a frame is inertial or not, we start by checking to see if the description of motion using that frame’s coordinates obeys the law of inertia: does an object left at rest on the counter in the laboratory stay at rest? If set in motion, does it move with constant velocity on a straight line? The Earth’s surface, as it turns out, is *not* quite a perfect inertial reference frame, but it is good enough that it made it possible for us to discover the law of inertia in the first place!

What spoils the inertial-ness of an Earth-bound reference frame is the Earth’s rotation, which, as we shall see later, is an example of *accelerated motion*. In fact, if you think about the grossly non-inertial frames I have introduced above—the bouncy plane, the braking car—they all have this in common: that their velocities are changing; they are *not* moving with constant speed on a straight line.

So, once you have found an inertial reference frame, to decide whether another one is inertial or not is simple: if it is moving with constant velocity (relative to the first, inertial frame), then it is itself inertial; if not, it is not. I will show you how this works, formally, in a little bit ([section 2.2](#)), after I (finally!) get around to properly introducing the concept of acceleration.

It is a fundamental principle of physics that *the laws of physics take the same form in all inertial reference frames*. The law of inertia is, of course, an example of such a law. Since all inertial frames are moving with constant velocity relative to each other, this

is another way to say that absolute motion is undetectable, and all motion is ultimately relative. Accordingly, this principle is known as the **principle of relativity**.

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2.2: Acceleration

Average and Instantaneous Acceleration

Just as we defined average velocity in the previous chapter, using the concept of displacement (or change in position) over a time interval Δt , we define *average acceleration* over the time Δt using the change in velocity:

$$a_{av} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i}. \quad (2.2.1)$$

Here, v_i and v_f are the initial and final velocities, respectively, that is to say, the velocities at the beginning and the end of the time interval Δt . As was the case with the average velocity, though, the average acceleration is a concept of somewhat limited usefulness, so we might as well proceed straight away to the definition of the *instantaneous acceleration* (or just “the” acceleration, without modifiers), through the same sort of limiting process by which we defined the instantaneous velocity:

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}. \quad (2.2.2)$$

Everything that we said in the previous chapter about the relationship between velocity and position can now be said about the relationship between acceleration and velocity. For instance (if you know calculus), the acceleration as a function of time is the derivative of the velocity as a function of time, which makes it the second derivative of the position function:

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2} \quad (2.2.3)$$

(and if you do *not* know calculus yet, do not worry about the superscripts “2” on that last expression! It is just a weird notation that you will learn someday.)

Similarly, we can “read off” the instantaneous acceleration from a *velocity* versus time graph, by looking at the slope of the line tangent to the curve at any point. However, if what we are given is a *position* versus time graph, the connection to the acceleration is more indirect. Figure 2.2.1 provides you with such an example. See if you can guess at what points along this curve the acceleration is positive, negative, or zero.

The way to do this “from scratch,” as it were, is to try to figure out what the velocity is doing, first, and infer the acceleration from that. Here is how that would go:

Starting at $t = 0$, and keeping an eye on the slope of the x -vs- t curve, we can see that the velocity starts at zero or near zero and increases steadily for a while, until t is a little bit more than 2 s (let us say, $t = 2.2$ s for definiteness). That would correspond to a period of positive acceleration, since Δv would be positive for every Δt in that range.

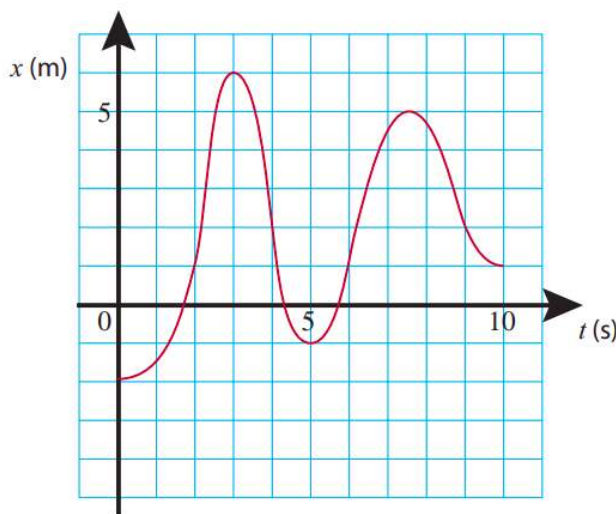


Figure 2.2.1: A possible position vs. time graph for an object whose acceleration changes with time.

Between $t = 2.2$ s and $t = 2.5$ s, as the object moves from $x = 2$ m to $x = 4$ m, the velocity does not appear to change very much, and the acceleration would correspondingly be zero or near zero. Then, around $t = 2.5$ s, the velocity starts to decrease noticeably,

becoming (instantaneously) zero at $t = 3$ s ($x = 6$ m). That would correspond to a negative acceleration. Note, however, that the velocity afterwards continues to decrease, becoming more and more negative until around $t = 4$ s. This also corresponds to a negative acceleration: even though the object is speeding up, it is speeding up in the negative direction, so Δv , and hence a , is negative for every time interval there. We conclude that $a < 0$ for all times between $t = 2.5$ s and $t = 4$ s.

Next, as we just look past $t = 4$ s, something else interesting happens: the object is still going in the negative direction (negative velocity), but now it is slowing down. Mathematically, that corresponds to a *positive* acceleration, since the algebraic value of the velocity is in fact increasing (a number like -3 is larger than a number like -5). Another way to think about it is that, if we have less and less of a negative thing, our overall trend is positive. So the acceleration is positive all the way from $t = 4$ s through $t = 5$ s (where the velocity is instantaneously zero as the object's direction of motion reverses), and beyond, until about $t = 6$ s, since between $t = 5$ s and $t = 6$ s the velocity is positive and growing.

You can probably figure out on your own now what happens after $t = 6$ s, reasoning as I did above, but you may also have noticed a pattern that makes this kind of analysis a lot easier. The acceleration (as those with a knowledge of calculus may have understood already), being proportional to the second derivative of the function $x(t)$ with respect to t , is directly related to the *curvature* of the x -vs- t graph. As figure 2.2.2 below shows, if the graph is *concave* (sometimes called “concave upwards”), the acceleration is positive, whereas it is negative whenever the graph is *convex* (or “concave downwards”). It is (instantly) zero at those points where the curvature changes (which you may know as *inflection points*), as well as over stretches of time when the x -vs- t graph is a straight line (motion with constant velocity).

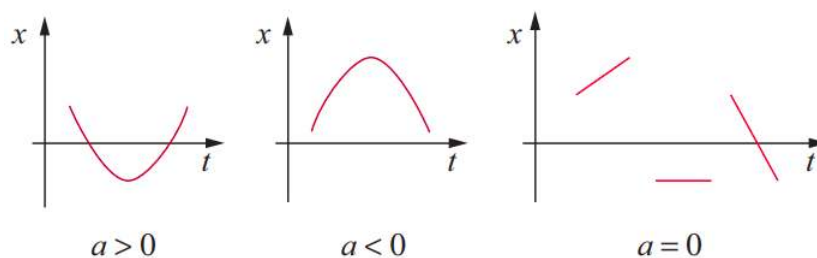


Figure 2.2.2: What the x -vs- t curves look like for the different possible signs of the acceleration.

Figure 2.2.3 shows position, velocity, and acceleration versus time for a hypothetical motion case. Please study it carefully until every feature of every graph makes sense, relative to the other two! You will see many other examples of this in the homework and the lab.

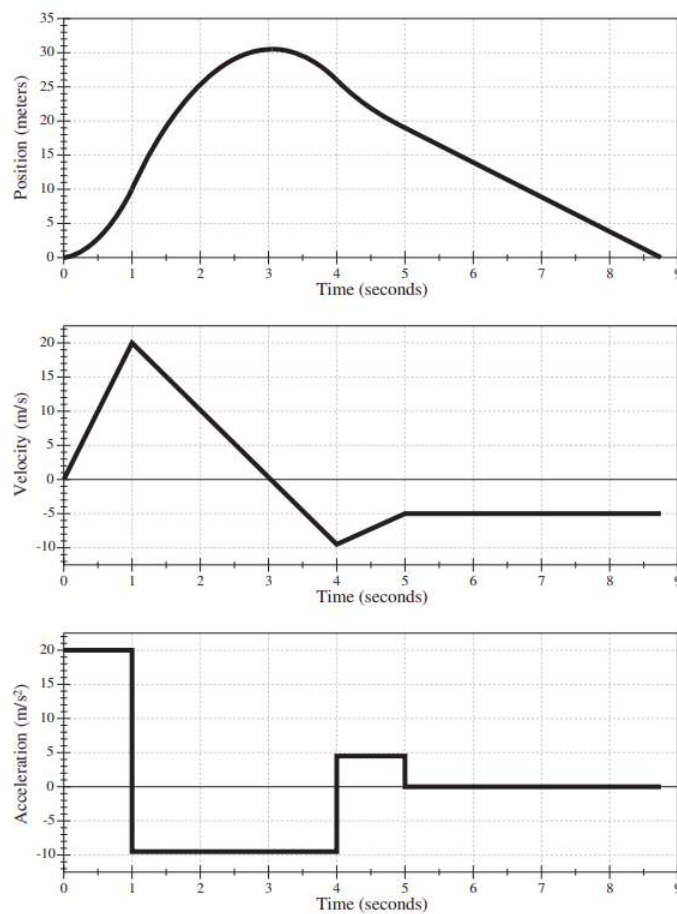


Figure 2.2.3: Sample position, velocity and acceleration vs. time graphs for motion with piecewise-constant acceleration.

Notice that, in all these figures, the sign of x or v at any given time has nothing to do with the sign of a at that same time. It is true that, for instance, a negative a , if sustained for a sufficiently long time, will eventually result in a negative v (as happens, for instance, in Figure 2.2.3 over the interval from $t = 1$ to $t = 4$ s) but this may take a long time, depending on the size of a and the initial value of v . The graphical clues to follow, instead, are: the acceleration is given by the slope of the tangent to the v -vs- t curve, or the curvature of the x -vs- t curve, as explained in Figure 2.2.2; and the velocity is given by the slope of the tangent to the x -vs- t curve.

(Note: To make the interpretation of Figure 2.2.3 simpler, I have chosen the acceleration to be “piecewise constant,” that is to say, constant over extended time intervals and changing in value discontinuously from one interval to the next. This is physically unrealistic: in any real-life situation, the acceleration would be expected to change more or less smoothly from instant to instant. We will see examples of that later on, when we start looking at realistic models of collisions.)

Motion With Constant Acceleration

A particular kind of motion that is both relatively simple and very important in practice is motion with constant acceleration (see Figure 2.2.3 again for examples). If a is constant, it means that the velocity changes with time at a constant rate, by a fixed number of m/s each second. (These are, incidentally, the units of acceleration: meters per second per second, or m/s^2 .) The change in velocity over a time interval Δt is then given by

$$\Delta v = a \Delta t \quad (2.2.4)$$

which can also be written

$$v = v_i + a(t - t_i). \quad (2.2.5)$$

Equation (2.2.5) is the form of the velocity function (v as a function of t) for motion with constant acceleration. This, in turn, has to be the derivative with respect to time of the corresponding position function. If you know simple derivatives, then, you can verify

that the appropriate form of the position function must be

$$x = x_i + v_i(t - t_i) + \frac{1}{2}a(t - t_i)^2 \quad (2.2.6)$$

or in terms of intervals,

$$\Delta x = v_i \Delta t + \frac{1}{2}a(\Delta t)^2. \quad (2.2.7)$$

Most often Equation (2.2.6) is written with the implicit assumption that the initial value of t is zero:

$$x = x_i + v_i t + \frac{1}{2}at^2. \quad (2.2.8)$$

This is simpler, but not as general as Equation (2.2.6). Always make sure that you know what conditions apply for any equation you decide to use!

As you can see from Equation (2.2.5), for intervals during which the acceleration is constant, the velocity vs. time curve should be a straight line. Figure 2.2.3 illustrates this. Equation (2.2.6), on the other hand, shows that for those same intervals the position vs. time curve should be a (portion of a) parabola, and again this can be seen in Figure 2.2.3 (sometimes, if the acceleration is small, the curvature of the graph may be hard to see; this happens in Figure 2.2.3 for the interval between $t = 4$ s and $t = 5$ s).

The observation that v -vs- t is a straight line when the acceleration is constant provides us with a simple way to derive Equation (2.2.7), when combined with the result (from the end of the previous chapter) that the displacement over a time interval Δt equals the area under the v -vs- t curve for that time interval. Indeed, consider the situation shown in Figure 2.2.4. The total area under the segment shown is equal to the area of a rectangle of base Δt and height v_i , plus the area of a triangle of base Δt and height $v_f - v_i$. Since $v_f - v_i = a\Delta t$, simple geometry immediately yields Equation (2.2.7), or its equivalent (2.2.6).

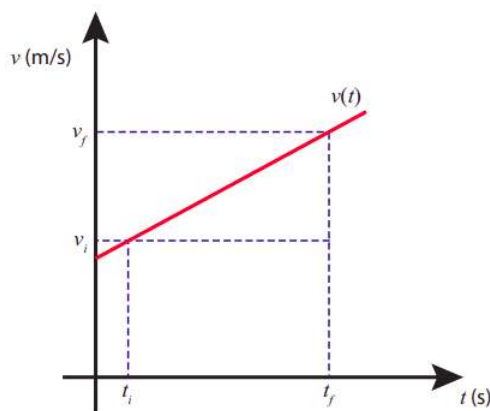


Figure 2.2.4: Graphical way to find the displacement for motion with constant acceleration.

Lastly, consider what happens if we solve Equation (2.2.4) for Δt and substitute the result in (2.2.7). We get

$$\Delta x = \frac{v_i \Delta v}{a} + \frac{(\Delta v)^2}{2a}. \quad (2.2.9)$$

Letting $\Delta v = v_f - v_i$, a little algebra yields

$$v_f^2 - v_i^2 = 2a\Delta x. \quad (2.2.10)$$

This is a handy little result that can also be seen to follow, more directly, from the work-energy theorems to be introduced in Chapter 7¹.

¹ In fact, equation (2.2.10) turns out to be so handy that you will probably find yourself using it over and over this semester, and you may even be tempted to use it for problems involving motion in two dimensions. However, unless you really know what you are doing, you should resist the temptation, since it is very easy to use Equation (2.2.10) incorrectly when the acceleration and the displacement do not lie along the same line. You should use the appropriate form of a work-energy theorem instead.

Acceleration as a Vector

In two (or more) dimensions we introduce the average acceleration vector

$$\vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t} = \frac{1}{\Delta t} (\vec{v}_f - \vec{v}_i) \quad (2.2.11)$$

whose components are $a_{av,x} = \Delta v_x / \Delta t$, etc.. The instantaneous acceleration is then the vector given by the limit of Equation (2.2.11) as $\Delta t \rightarrow 0$, and its components are, therefore, $a_x = dv_x/dt$, $a_y = dv_y/dt$, . . .

Note that, since $\vec{v}_i$ and $\vec{v}_f$ in Equation (2.2.11) are vectors, and have to be subtracted as such, the acceleration vector will be nonzero whenever $\vec{v}_i$ and $\vec{v}_f$ are different, even if, for instance, their magnitudes (which are equal to the object's speed) are the same. In other words, you have accelerated motion whenever the *direction* of motion changes, even if the speed does not.

As long as we are working in one dimension, I will follow the same convention for the acceleration as the one I introduced for the velocity in [Chapter 1](#): namely, I will use the symbol a , without a subscript, to refer to the relevant component of the acceleration ($a_x, a_y, \dots$), and *not* to the magnitude of the vector $\vec{a}$.

Acceleration in Different Reference Frames

In [Chapter 1](#) you saw that the following relation holds between the velocities of a particle P measured in two different reference frames, A and B:

$$\vec{v}_{AP} = \vec{v}_{AB} + \vec{v}_{BP}. \quad (2.2.12)$$

What about the acceleration? An equation like (2.2.12) will hold for the initial and final velocities, and subtracting them we will get

$$\Delta \vec{v}_{AP} = \Delta \vec{v}_{AB} + \Delta \vec{v}_{BP}. \quad (2.2.13)$$

Now suppose that reference frame B moves with *constant velocity* relative to frame A. In that case, $\vec{v}_{AB,f} = \vec{v}_{AB,i}$, so $\Delta \vec{v}_{AB} = 0$, and then, dividing Equation (2.2.13) by Δt , and taking the limit $\Delta t \rightarrow 0$, we get

$$\vec{a}_{AP} = \vec{a}_{BP} \quad (\text{for constant } \vec{v}_{AB}). \quad (2.2.14)$$

So, if two reference frames are moving at constant velocity relative to each other, observers in both frames measure the *same* acceleration for any object they might both be tracking.

The result Equation (2.2.14) means, in particular, that if we have an inertial frame then any frame moving at constant velocity relative to it will be inertial too, since the respective observers' measurements will agree that an object's velocity does not change (otherwise put, its acceleration is zero) when no forces act on it. Conversely, an accelerated frame will *not* be an inertial frame, because Equation (2.2.14) will not hold. This is consistent with the examples I mentioned in [Section 2.1](#) (the bouncing plane, the car coming to a stop). Another example of a non-inertial frame would be a car going around a curve, even if it is going at constant speed, since, as I just pointed out above, this is also an accelerated system. This is confirmed by the fact that objects in such a car tend to move—relative to the car—towards the outside of the curve, even though no actual force is acting on them.

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2.3: Free Fall

An important example of motion with (approximately) constant acceleration is provided by *free fall* near the surface of the Earth. We say that an object is in “free fall” when the only force acting on it is the force of gravity (the word “fall” here may be a bit misleading, since the object could actually be moving upwards some of the time, if it has been thrown straight up, for instance). The space station is in free fall, but because it is nowhere near the surface of the earth its direction of motion (and hence its acceleration, regarded as a two-dimensional vector) is constantly changing. Right next to the surface of the earth, on the other hand, the planet’s curvature is pretty much negligible and gravity provides an approximately constant, vertical acceleration, which, in the absence of other forces, turns out to be *the same for every object*, regardless of its size, shape, or weight.

The above result—that, in the absence of other forces, all objects should fall to the earth at the same rate, regardless of how big or heavy they are—is so contrary to our common experience that it took many centuries to discover it. The key, of course, as with the law of inertia, is to realize that, under normal circumstances, frictional forces are, in fact, acting all the time, so an object falling through the atmosphere is never *really* in “free” fall: there is always, at a minimum, and in addition to the force of gravity, an air drag force that opposes its motion. The magnitude of this force does depend on the object’s size and shape (basically, on how “aerodynamic” the object is); and thus a golf ball, for instance, falls much faster than a flat sheet of paper. Yet, if you crumple up the sheet of paper till it has the same size and shape as the golf ball, you can see for yourself that they do fall at approximately the same rate! The equality can never be exact, however, unless you get rid completely of air drag, either by doing the experiment in an evacuated tube, or (in a somewhat extreme way), by doing it on the surface of the moon, as the Apollo 15 astronauts did with a hammer and a feather back in 1971².

This still leaves us with something of a mystery, however: the force of gravity is the only force known to have the property that it imparts all objects the *same* acceleration, regardless of their mass or constitution. A way to put this technically is that the force of gravity on an object is proportional to that object’s *inertial mass*, a quantity that we will introduce properly in the next chapter. For the time being, we will simply record here that this acceleration, near the surface of the earth, has a magnitude of approximately 9.8 m/s^2 , a quantity that is denoted by the symbol g . Thus, if we take the upwards direction as positive (as is usually done), we get for the acceleration of an object in free fall $a = -g$, and the equations of motion become

$$\Delta v = -g\Delta t \quad (2.3.1)$$

$$\Delta y = v_i\Delta t - \frac{1}{2}g(\Delta t)^2 \quad (2.3.2)$$

where I have used y instead of x for the position coordinate, since that is a more common choice for a vertical axis. Note that we could as well have chosen the downward direction as positive, and that may be a more natural choice in some problems. Regardless, the quantity g is always defined to be positive: $g = 9.8 \text{ m/s}^2$. The acceleration, then, is g or $-g$, depending on which direction we take to be positive

In practice, the value of g changes a little from place to place around the earth, for various reasons (it is somewhat sensitive to the density of the ground below you, and it decreases as you climb higher away from the center of the earth). In a later chapter we will see how to calculate the value of g from the mass and radius of the earth, and also how to calculate the equivalent quantity for other planets.

In the meantime, we can use equations like (2.3.1) and (2.3.2) (as well as (2.2.10), with the appropriate substitutions) to answer a number of interesting questions about objects thrown or dropped straight up or down (always, of course, assuming that air drag is negligible). For instance, back at the beginning of this chapter I mentioned that if I dropped an object it might take about half a second to hit the ground. If you use Equation (2.3.2) with $v_i = 0$ (since I am dropping the object, not throwing it down, its initial velocity is zero), and substitute $\Delta t = 0.5 \text{ s}$, you get $\Delta y = 1.23 \text{ m}$ (about 4 feet). This is a reasonable height from which to drop something.

On the other hand, you may note that half a second is not a very long time in which to make accurate observations (especially if you do not have modern electronic equipment), and as a result of that there was considerable confusion for many centuries as to the precise way in which objects fell. Some people believed that the speed did increase in some way as the object fell, while others appear to have believed that an object dropped would “instantaneously” (that is, at soon as it left your hand) acquire some speed and keep that unchanged all the way down. In reality, in the presence of air drag, what happens is a combination of both: initially the speed increases at an approximately constant rate (free, or nearly free fall), but the drag force increases with the speed as well, until eventually it balances out the force of gravity, and from that point on the speed does not increase anymore: we say that the

object has reached “terminal velocity.” Some objects reach terminal velocity almost instantly, whereas others (the more “aerodynamic” ones) may take a long time to do so. This accounts for the confusion that prevailed before Galileo’s experiments in the early 1600’s.

Galileo’s main insight, on the theoretical side, was the realization that it was necessary to separate clearly the effect of gravity and the effect of the drag force. Experimentally, his big idea was to use an inclined plane to slow down the “fall” of an object, so as to make accurate measurements possible (and also, incidentally, reduce the air drag force!). These “inclined planes” were just basically ramps down which he sent small balls (like marbles) rolling. By changing the steepness of the ramp he could control how slowly the balls moved. He reasoned that, ultimately, the force that made the balls go down was essentially the same force of gravity, only not the whole force, but just a fraction of it. Today we know that, in fact, an object sliding (*not* rolling!) up or down on a *frictionless* incline will experience an acceleration directed downwards along the incline and with a magnitude equal to $g \sin \theta$, where θ is the angle that the slope makes with the horizontal:

$$a = g \sin \theta \quad (\text{inclined plane, taking } \textit{downwards} \text{ to be positive}) \quad (2.3.3)$$

(for some reason, it seems more natural, when dealing with inclined planes, to take the downward direction as positive!). Equation (2.3.3) makes sense in the two extreme cases in which the plane is completely vertical ($\theta = 90^\circ$, $a = g$) and completely horizontal ($\theta = 0^\circ$, $a = 0$). For intermediate values, you will carry out experiments in the lab to verify this result.

We will show, in a later chapter, how Equation (2.3.3) comes about from a careful consideration of all the forces acting on the object; we will also see, later on, how it needs to be modified for the case of a rolling, rather than a sliding, object. This modification does not affect Galileo’s main conclusion, which was, basically, that the natural falling motion in the absence of friction or drag forces is motion with *constant acceleration* (at least, near the surface of the earth, where g is constant to a very good approximation).

²The video of this is available online: <https://www.youtube.com/watch?v=oYEgdZ3iEKA>. It is, however, pretty low resolution and hard to see. A very impressive modern-day demonstration involving feathers and a bowling ball in a completely evacuated (airless) room is available here: <https://www.youtube.com/watch?v=E43-CfukEgs>.

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2.4: In Summary

1. The *law of inertia* states that, if no external influences (forces) are acting on an object, then, if the object is initially at rest it will stay at rest, and if it is initially moving it will continue to move with constant velocity (unchanging speed and direction).
2. Reference frames in which the law of inertia is seen to hold (when the velocities of objects are calculated from their coordinates in that frame) are called *inertial*. A reference frame that is moving at constant velocity relative to an inertial frame is also an inertial frame. Conversely, accelerated reference frames are non-inertial.
3. Motion with constant velocity is fundamentally indistinguishable from no motion at all (i.e., rest). As long as the velocity (of the objects involved) does not change, only *relative* motion can be detected. This is known as the **principle of relativity**. Another way to state it is that the laws of physics must take the same form in all inertial reference frames (so you cannot single out one as being in “absolute rest” or “absolute motion”).
4. *Changes* in velocity are detectable, and, by (1) above, are evidence of unbalanced forces acting on an object.
5. The rate of change of an object’s velocity is the object’s *acceleration*: the average acceleration over a time interval Δt is $a_{av} = \Delta v / \Delta t$, and the instantaneous acceleration at a time t is the limit of the average acceleration calculated for successively shorter time intervals Δt , all with the same initial time $t_i = t$. Mathematically, this means the acceleration is the derivative of the velocity function, $a = dv/dt$.
6. In a velocity versus time graph, the acceleration can be read from the slope of the line tangent to the curve (just like the velocity in a position versus time graph).
7. In a position versus time graph, the regions with positive acceleration correspond to a concave curvature (like a parabola opening up), and those with negative acceleration correspond to a convex curvature (like a parabola opening down). Points of inflection (where the curvature changes) and straight lines correspond to points where the acceleration is zero.
8. The basic equations used to describe motion with constant acceleration are (2.2.4), (2.2.7) and (2.2.10) above. Alternative forms of these are also provided in the text.
9. In more than one dimension, a change in the *direction* of the velocity vector results in a nonzero acceleration, even if the object’s speed does not change.
10. An object is said to be in *free fall* when the only force acting on it is gravity. All objects in free fall experience the same acceleration at the same point in their motion, regardless of their mass or composition. Near the surface of the earth, this acceleration is approximately constant and has a magnitude $g = 9.8 \text{ m/s}^2$.
11. An object sliding on a frictionless inclined plane experiences (if air drag is negligible) an acceleration directed downward along the incline and with a magnitude $g \sin \theta$, where θ is the angle the incline makes with the horizontal.

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2.5: Examples

Example 2.5.1: Motion with piecewise constant acceleration

Construct the position vs. time, velocity vs. time, and acceleration vs. time graphs for the motion described below. For each of the intervals (a)–(d) you'll need to figure out the position (height) and velocity of the rocket at the beginning and the end of the interval, and the acceleration for the interval. In addition, for interval (b) you need to figure out the maximum height reached by the rocket and the time at which it occurs. For interval (d) you need to figure out its duration, that is to say, the time at which the rocket hits the ground.

- A rocket is shot upwards, accelerating from rest to a final velocity of 20 m/s in 1 s as it burns its fuel. (Treat the acceleration as constant during this interval.)
- From $t = 1$ s to $t = 4$ s, with the fuel exhausted, the rocket flies under the influence of gravity alone. At some point during this time interval (you need to figure out when!) it stops climbing and starts falling.
- At $t = 4$ s a parachute opens, suddenly causing an upwards acceleration (again, treat it as constant) lasting 1 s; at the end of this interval, the rocket's velocity is 5 m/s downwards.
- The last part of the motion, with the parachute deployed, is with constant velocity of 5 m/s downwards until the rocket hits the ground.

Solution

(a) For this first interval (for which I will use a subscript "1" throughout) we have

$$\Delta y_1 = \frac{1}{2} a_1 (\Delta t_1)^2 \quad (2.5.1)$$

using Equation (2.2.6) for motion with constant acceleration with zero initial velocity (I am using the variable y , instead of x , for the vertical coordinate; this is more or less customary, but, of course, I could have used x just as well).

Since the acceleration is constant, it is equal to its average value:

$$a_1 = \frac{\Delta v}{\Delta t} = 20 \frac{\text{m}}{\text{s}^2}.$$

Substituting this into (2.5.1) we get the height at $t = 1$ s is 10 m. The velocity at that time, of course, is $v_{f1} = 20$ m/s, as we were told in the statement of the problem.

(b) This part is free fall with initial velocity $v_{i2} = 20$ m/s. To find how high the rocket climbs, use Equation (2.3.1) in the form $v_{top} - v_{i2} = -g(t_{top} - t_{i2})$, with $v_{top} = 0$ (as the rocket climbs, its velocity decreases, and it stops climbing when its velocity is zero). This gives us $t_{top} = 3.04$ s as the time at which the rocket reaches the top of its trajectory, and then starts coming down. The corresponding displacement is, by Equation (2.3.2),

$$\Delta y_{top} = v_{i2} (t_{top} - t_{i2}) - \frac{1}{2} g (t_{top} - t_{i2})^2 = 20.4 \text{ m}$$

so the maximum height it reaches is 30.4 m.

At the end of the full 3-second interval, the rocket's displacement is

$$\Delta y_2 = v_{i2} \Delta t_2 - \frac{1}{2} g (\Delta t_2)^2 = 15.9 \text{ m}$$

(so its height is 25.9 m above the ground), and the final velocity is

$$v_{f2} = v_{i2} - g \Delta t_2 = -9.43 \frac{\text{m}}{\text{s}}.$$

(c) The acceleration for this part is $(v_{f3} - v_{i3})/\Delta t_3 = (-5 + 9.43)/1 = 4.43 \text{ m/s}^2$. Note the positive sign. The displacement is

$$\Delta y_3 = -9.43 \times 1 + \frac{1}{2} \times 4.43 \times 1^2 = -7.22 \text{ m}$$

so the final height is $25.9 - 7.21 = 18.7$ m.

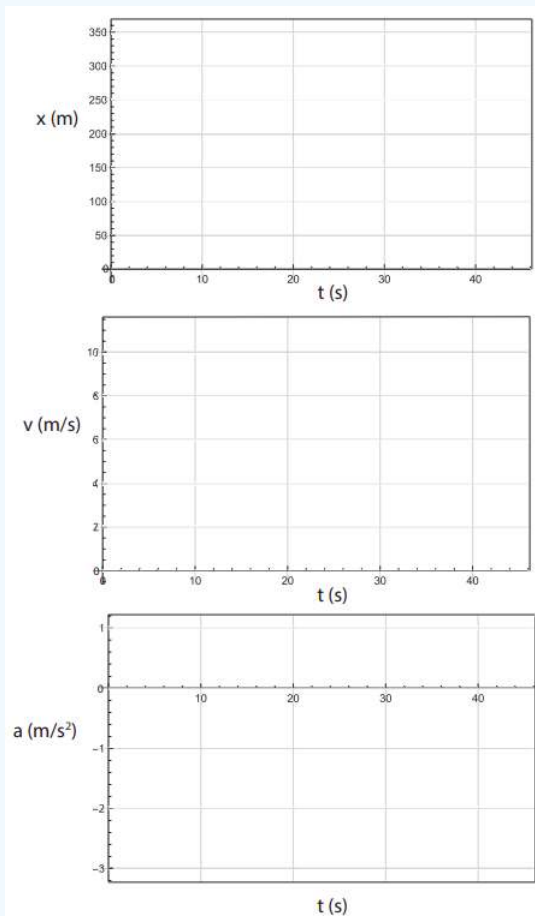
(d) This is just motion with constant speed to cover 18.7 m at 5 m/s. The time it takes is 3.74 s. The graphs for this motion are shown earlier in the chapter, in [Figure 2.2.3](#).

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2.6: Exercises

Exercise 2.6.1

You get on your bicycle and ride it with a constant acceleration of 0.5 m/s^2 for 20 s. After that, you continue riding at a constant velocity for a distance of 200 m. Finally, you slow to a stop, with a constant acceleration, over a distance of 20 m.



- How far did you travel while you were accelerating at 0.5 m/s^2 , and what was your velocity at the end of that interval?
- After that, how long did it take you to cover the next 200 m?
- What was your acceleration while you were slowing down to a stop, and how long did it take you to come to a stop?
- Considering the whole trip, what was your average velocity?
- Plot the position versus time, velocity versus time, and acceleration versus time graphs for the whole trip, in the grids provided above. Values at the beginning and end of each interval must be exact. Slopes and curvatures must be represented accurately. Do not draw any of the curves beyond the time the rider stops (or, if you do, make sure what you draw makes sense!).

Exercise 2.6.2

You throw an object straight upwards and catch it again, when it comes down to the same initial height, 2 s later.

- How high did it rise above its initial height?
- With what initial speed did you throw it?

(Note: for this problem you should use the fact that, if air drag is negligible, the object will return to its initial height with the same speed it had initially.)

Exercise 2.6.3

You are trying to catch up with a car that is in front of you on the highway. Initially you are both moving at 25 m/s, and the distance between you is 100 m. You step on the gas and sustain a constant acceleration for a time $\Delta t = 30$ s, at which point you have pulled even with the other car.

- What is 25 m/s, in miles per hour?
- What was your acceleration over the 30 s time interval?
- How fast were you going when you caught up with the other car?

Exercise 2.6.4

Go back to Exercise 1.6.4 of Chapter 1, and use the information in the figure to draw an accurate position vs. time graph for both runners.

Exercise 2.6.5

A child on a sled slides (starting from rest) down an icy slope that makes an angle of 15° with the horizontal. After sliding 20 m down the slope, the child enters a flat, slushy region, where she slides for 2.0 s with a constant negative acceleration of -1.5 m/s^2 with respect to her direction of motion. She then slides up another icy slope that makes a 20° angle with the horizontal.

- How fast was the child going when she reached the bottom of the first slope? How long did it take her to get there?
- How long was the flat stretch at the bottom?
- How fast was the child going as she started up the second slope?
- How far up the second slope did she slide?

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CHAPTER OVERVIEW

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University of Arkansas

3: Momentum and Inertia

Julio Gea-Banacloche

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3.1: Inertia

In everyday language, we speak of something or someone “having a large inertia” to mean, essentially, that they are very difficult to set in motion. This usage of the word “inertia” is consistent with the “law of inertia” we introduced in the previous chapter (which states, among other things, that an object at rest, if left to itself, will just remain at rest), but it goes a bit beyond that by trying to quantify just how hard it may be to get the object to move.

We do know, from experience, that lighter objects are easier to set in motion than heavier objects, but most of us probably have an intuition that gravity (the force that pulls an object towards the earth and hence determines its weight) is not involved in an essential way here. Imagine, for instance, the difference between slapping a volleyball and a bowling ball. It is not hard to believe that the latter would hurt as much if we did it while floating in free fall in the space station (in a state of effective “weightlessness”) as if we did it right here on the surface of the earth. In other words, it is not (necessarily) how heavy something feels, but just how *massive* it is.

But just what is this “massiveness” quality that we associate intuitively with a large inertia? Is there a way (other than resorting to the weight again) to assign to it a numerical value?

Relative Inertia and Collisions

One possible way to determine the *relative* inertias of two objects, conceptually, at least, is to try to use one of them to set the other one in motion. Most of us are familiar with what happens when two identical objects (presumably, therefore, having the same inertia) collide: if the collision is head-on (so the motion, before and after, is confined to a straight line), they basically exchange velocities. For instance, a billiard ball hitting another one will stop dead and the second one will set off with the same speed as the first one. The toy sometimes called “Newton’s balls” or “Newton’s cradle” also shows this effect. Intuitively, we understand that what it takes to stop the first ball is exactly the same as it would take to set the second one in motion with the same velocity.

But what if the objects colliding have different inertias? We expect that the change in their velocities as a result of the collision will be different: the velocity of the object with the largest inertia will not change very much, and conversely, the change in the velocity of the object with the smallest inertia will be comparatively larger. A velocity vs. time graph for the two objects might look somewhat like the one sketched in Figure 3.1.1.

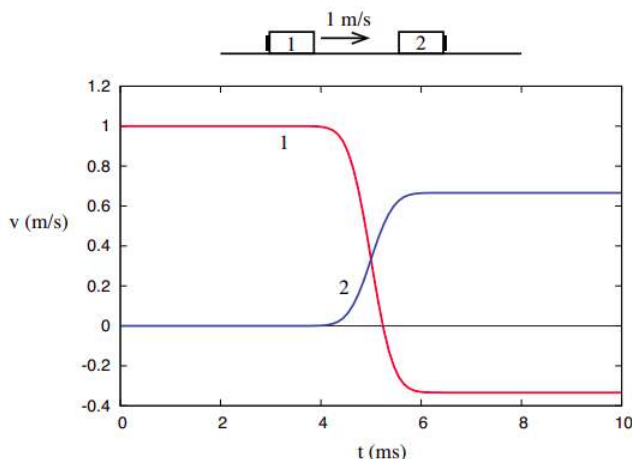


Figure 3.1.1: An example of a velocity vs. time graph for a collision of two objects with different inertias.

In this picture, object 1, initially moving with velocity $v_{1i} = 1$ m/s, collides with object 2, initially at rest. After the collision, which here is assumed to take a millisecond or so, object 1 actually bounces back, so its final velocity is $v_{1f} = -1/3$ m/s, whereas object 2 ends up moving to the right with velocity $v_{2f} = 2/3$ m/s. So the change in the velocity of object 1 is $\Delta v_1 = v_{1f} - v_{1i} = -4/3$ m/s, whereas for object 2 we have $\Delta v_2 = v_{2f} - v_{2i} = 2/3$ m/s.

It is tempting to use this ratio, $\Delta v_1/\Delta v_2$, as a measure of the *relative inertia* of the two objects, only we’d want to use it upside down and with the opposite sign: that is, since $\Delta v_2/\Delta v_1 = -1/2$ we would say that object 2 has *twice* the inertia of object 1. But then we have to ask: is this a reliable, repeatable measure? Will it work for any kind of collision (within reason, of course: we clearly need to stay in one dimension, and eliminate external influences such as friction), and for any initial velocity?

To begin with, we have reason to expect that it does not matter whether we shoot object 1 towards object 2 or object 2 towards object 1, because we learned in the previous chapter that *only relative motion is detectable*, and the relative motion is the same in both cases. Consider, for instance, what the collision in Figure 3.1.1 appears like to a hypothetical observer moving along with object 1, at 1 m/s. To him, object 1 appears to be at rest, and it is object 2 that is coming towards him, with a velocity of -1 m/s. To see what the outcome of the collision looks like to him, just add the same -1 m/s to the final velocities we obtained before: object 1 will end up moving at $v_{1f} = -4/3$ m/s, and object 2 would move at $v_{2f} = -1/3$ m/s, and we would have a situation like the one shown in Figure 3.1.2, where both curves have simply been shifted down by 1 m/s:

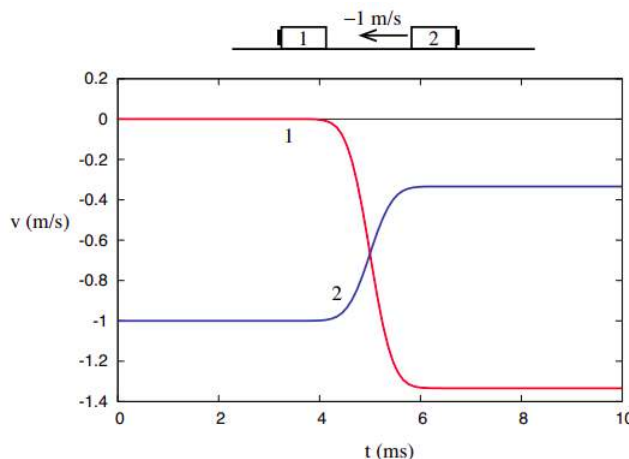


Figure 3.1.2: Another example (really the same collision as in Figure 3.1.1, only as seen by an observer initially moving to the right at 1 m/s).

But then, this is exactly what we should expect to find also in our laboratory if we actually did send the second object at 1 m/s towards the first one sitting at rest. All the individual velocities have changed relative to Figure 3.1.1, but the *velocity changes*, Δv_1 and Δv_2 , are clearly still the same, and therefore so is our (tentative) measure of the objects' relative inertia.

Clearly, the same argument can be used to conclude that the same result will be obtained when both objects are initially moving towards each other, as long as their *relative velocity* is the same as in these examples, namely, 1 m/s. However, unless we do the experiments we cannot really predict what will happen if we increase (or decrease) their relative velocity. In fact, we could imagine smashing the two objects at very high speed, so they might even become seriously mangled in the process. Yet, experimentally (and this is not at all an obvious result!), we would still find the same value of $-1/2$ for the ratio $\Delta v_2/\Delta v_1$, at least as long as the collision is not so violent that the objects actually break up into pieces.

Perhaps the most surprising result of our experiments would be the following: imagine that the objects have a “sticky” side (for instance, the small black rectangles shown in the pictures could be strips of Velcro), and we turn them around so that when they collide they will end up stuck to each other. In this case (which, as we shall see later, is termed a **completely inelastic** collision), the v -vs- t graph might look like Figure 3.1.3 below.

Now the two objects end up moving together to the right, fairly slowly: $v_{1f} = v_{2f} = 1/3$ m/s. The velocity changes are $\Delta v_1 = -2/3$ m/s and $\Delta v_2 = 1/3$ m/s, both of which are different from what they were before, in Figs. 3.1.1 and 3.1.2: yet, the ratio $\Delta v_2/\Delta v_1$ is still equal to $-1/2$, just as in all the previous cases.

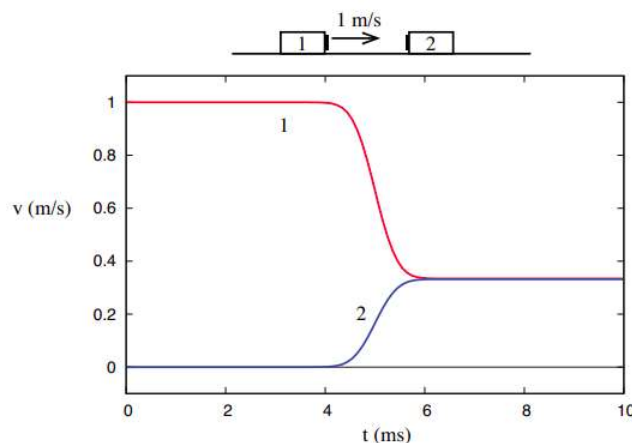


Figure 3.1.1 became stuck together when they collided.

Inertial Mass: Definition and Properties

At this point, it would seem reasonable to assume that this ratio, $\Delta v_2/\Delta v_1$, is, in fact, telling us something about an *intrinsic* property of the two objects, what we have called above their “relative inertia.” It is easy, then, to see how one could assign a value to the inertia of any object (at least, conceptually): choose a “standard” object, and decide, arbitrarily, that its inertia will have the numerical value of 1, in whichever units you choose for it (these units will turn out, in fact, to be kilograms, as you will see in a minute). Then, to determine the inertia of another object, which we will label with the subscript 1, just arrange a one-dimensional collision between object 1 and the standard, under the right conditions (basically, no net external forces), measure the velocity changes Δv_1 and Δv_s , and take the quantity $-\Delta v_s/\Delta v_1$ as the numerical value of the ratio of the inertia of object 1 to the inertia of the standard object. In symbols, using the letter m to represent an object’s inertia,

$$\frac{m_1}{m_s} = -\frac{\Delta v_s}{\Delta v_1} \quad (3.1.1)$$

But, since $m_s = 1$ by definition, this gives us directly the numerical value of m_1 .

The reason we use the letter m is, as you must have guessed, because, in fact, the inertia defined in this way turns out to be identical to what we have traditionally called “mass.” More precisely, the quantity defined this way is an object’s *inertial mass*. The remarkable fact, mentioned earlier, that the force of gravity between two objects turns out to be proportional to their inertial masses, allows us to determine the inertial mass of an object by the more traditional procedure of simply weighing it, rather than elaborately staging a collision between it and the standard kilogram on an ice-hockey rink. But, in principle, we could conceive of the existence of two different quantities that should be called “inertial mass” and “gravitational mass,” and the identity (or more precisely, the—so far as we know—exact proportionality) of the two is a rather mysterious experimental fact¹.

In any case, by the way we have constructed it, the inertial mass, defined as in Equation (3.1.1), does capture, in a quantitative way, the concept that we were trying to express at the beginning of the chapter: namely, how difficult it may be to set an object in motion. In principle, however, other experiments would need to be conducted to make sure that it does have the properties we have traditionally associated with the concept of mass. For instance, suppose we join together two objects of mass m . Is the mass of the resulting object $2m$? Collision experiments would, indeed, show this to be the case with great accuracy in the macroscopic world (with which we are concerned this semester), but this is a good example of how you cannot take anything for granted: at the microscopic level, it is again a fact that the inertial mass of an atomic nucleus is a little *less* than the sum of the masses of all its constituent protons and neutrons².

Probably the last thing that would need to be checked is that *the ratio of inertias is independent of the standard*. Suppose that we have two objects, to which we have assigned masses m_1 and m_2 by arranging for each to collide with the “standard object” independently. If we now arrange for a collision between objects 1 and 2 directly, will we actually find that the ratio of their velocity changes is given by the ratio of the separately determined masses m_1 and m_2 ? We certainly would need that to be the case, in order for the concept of inertia to be truly useful; but again, we should not assume anything until we have tested it! Fortunately, the tests would indeed reveal that, in every case, the expected relationship holds³

$$-\frac{\Delta v_2}{\Delta v_1} = \frac{m_1}{m_2}. \quad (3.1.2)$$

At this point, we are not just in possession of a useful definition of inertia, but also of a veritable *law of nature*, as I will explain next.

¹This fact, elevated to the category of a principle by Einstein (the *equivalence principle*) is the starting point of the general theory of relativity.

²And this is not just an unimportant bit of trivia: all of nuclear power depends on this small difference.

³Equation 3.1.2 actually is found to hold also at the microscopic (or *quantum*) level, although there we prefer to state the result by saying that conservation of momentum holds (see the following section).

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3.2: Momentum

For an object of (inertial) mass m moving, in one dimension, with velocity v , we define its *momentum* as

$$p = mv \quad (3.2.1)$$

(the choice of the letter p for momentum is apparently related to the Latin word “impetus”).

We can think of momentum as a sort of extension of the concept of inertia, from an object at rest to an object in motion. When we speak of an object’s inertia, we typically think about what it may take to get it moving; when we speak of its momentum, we typically think of that it may take to stop it (or perhaps deflect it). So, both the inertial mass m and the velocity v are involved in the definition.

We may also observe that what looks like inertia in some reference frame may look like momentum in another. For instance, if you are driving in a car towing a trailer behind you, the trailer has only a large amount of inertia, but no momentum, relative to you, because its velocity relative to you is zero; however, the trailer definitely has a large amount of momentum (by virtue of both its inertial mass and its velocity) relative to somebody standing by the side of the road.

Conservation of Momentum; Isolated Systems

For a system of objects, we treat the momentum as an *additive* quantity. So, if two colliding objects, of masses m_1 and m_2 , have initial velocities v_{1i} and v_{2i} , we say that the total initial momentum of the system is $p_i = m_1 v_{1i} + m_2 v_{2i}$, and similarly if the final velocities are v_{1f} and v_{2f} , the total final momentum will be $p_f = m_1 v_{1f} + m_2 v_{2f}$.

We then assert that *the total momentum of the system is not changed by the collision*. Mathematically, this means

$$p_i = p_f \quad (3.2.2)$$

or

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}. \quad (3.2.3)$$

But this last equation, in fact, follows directly from Equation (3.1.2): to see this, move all the quantities in Equation (3.2.3) having to do with object 1 to one side of the equal sign, and those having to do with object 2 to the other side. You then get

$$\begin{aligned} m_1 (v_{1i} - v_{1f}) &= m_2 (v_{2f} - v_{2i}) \\ -m_1 \Delta v_1 &= m_2 \Delta v_2 \end{aligned} \quad (3.2.4)$$

which is just another way to write Equation (3.1.2). Hence, the result (3.1.2) ensures the conservation of the total momentum of a system of any two interacting objects (“particles”), regardless of the form the interaction takes, as long as there are no external forces acting on them.

Momentum conservation is one of the most important principles in all of physics, so let me take a little time to explain how we got here and elaborate on this result. First, as I just mentioned, we have been more or less implicitly assuming that the two interacting objects form an *isolated* system, by which we mean that, throughout, they interact with nothing other than each other. (Equivalently, there are no external forces acting on them.)

It is pretty much impossible to set up a system so that it is *really* isolated in this strict sense; instead, in practice, we settle for making sure that the external forces on the two objects *cancel out*. This is what happens on the air tracks with which you will be doing experiments this semester: gravity is acting on the carts, but that force is balanced out by the upwards push of the air from the track. A system on which there is no *net* external force is as good as isolated for practical purposes, and we will refer to it as such. (It is harder, of course, to completely eliminate friction and drag forces, so we just have to settle for approximately isolated systems in practice.)

Secondly, we have assumed so far that the motion of the two objects is restricted to a straight line—one dimension. In fact, momentum is a *vector* quantity (just like velocity is), so in general we should write

$$\vec{p} = m\vec{v}$$

and conservation of momentum, in general, holds as a vector equation for any isolated system in three dimensions:

$$\vec{p}_i = \vec{p}_f. \quad (3.2.5)$$

What this means, in turn, is that each separate component (x , y and z) of the momentum will be separately conserved (so Equation (3.2.5) is equivalent to three scalar equations, in three dimensions). When we get to study the vector nature of forces, we will see an interesting implication of this, namely, that it is possible for one component of the momentum vector to be conserved, but not another—depending on whether there is or there isn't a net external force in that direction or not. For example, anticipating things a bit, when you throw an object horizontally, as long as you can ignore air drag, there is no horizontal force acting on it, and so that component of the momentum vector is conserved, but the vertical component is changing all the time because of the (vertical) force of gravity.

Thirdly, although this may not be immediately obvious, for an isolated system of two colliding objects the momentum is truly conserved throughout the whole collision process. It is not just a matter of comparing the initial and final velocities: at any of the times shown in Figures 3.1.1 through 3.1.3, if we were to measure v_1 and v_2 and compute $m_1v_1 + m_2v_2$, we would obtain the same result. In other words, the total momentum of an isolated system is *constant*: it has the same value at all times.

Finally, all these examples have involved interactions between only two particles. Can we really generalize this to conclude that the total momentum of an isolated system of any number of particles is constant, even when all the particles may be interacting with each other simultaneously? Here, again, the experimental evidence is overwhelmingly in favor of this hypothesis⁴, but much of our confidence on its validity comes in fact from a consideration of the nature of the internal interactions themselves. It is a mathematical fact that all of the interactions so far known to physics have the property of conserving momentum, whether acting individually or simultaneously. No experiments have ever suggested the existence of an interaction that does not have this property.

⁴For an important piece of indirect evidence, just consider that any extended object is in reality a collection of interacting particles, and the experiments establishing conservation of momentum almost always involve such extended objects. See the following section for further thoughts on this matter.

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